TAXATION AND WELFARE IN GHANA: AN AUTOREGRESSIVE DISTRIBUTED LAG (ARDL) APPROACH

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Declaration

I, XXXXXXX, affirm that this thesis is the result of my own work and has not been presented to any other institution.

Signature:

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Approval

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Abstract

Background: Taxation is a critical instrument for government financing and socio-economic development, funding essential public services including education, healthcare, and social protection programs. However, concerns persist regarding the extent to which taxes influence the welfare of Ghanaians, particularly among vulnerable populations. This study examined the impact of taxation on welfare in Ghana.

Method: The study adopted a quantitative research design using annual time series data for Ghana from 1983 to 2022 (40 observations; 38 observations, 1985–2022, after allowing for two lags) sourced from the Ghana Statistical Service, the World Bank World Development Indicators, and the UNU-WIDER Government Revenue Dataset. Tax variables were measured as a percentage of GDP. Missing years in the survey-based Gini coefficient and poverty headcount (observed in only seven Ghana Living Standards Survey years) and in secondary school enrollment were filled by linear interpolation. Stationarity was examined using the Augmented Dickey-Fuller (ADF) unit root test, and the Autoregressive Distributed Lag (ARDL) bounds testing approach to cointegration was employed to estimate the short-run and long-run effects of direct, indirect, and total taxes on income inequality, secondary school enrollment, immunization coverage (access to healthcare), and poverty headcount.

Results: Taxes showed no significant effect on income inequality in either the short run or the long run, indicating that Ghana's tax system functions primarily for revenue generation rather than redistribution. For education, direct taxes had a significant delayed positive effect: a one percentage point increase in direct taxes (% of GDP) was associated with a 4.60 percentage point rise in secondary school enrollment two years later. For healthcare, a stable long-run relationship was found, with indirect taxes significantly raising immunization coverage in the

long run and about 38% of any deviation from equilibrium corrected each year. For poverty, direct taxes had no significant effect, while government expenditure was associated with lower poverty; because poverty data are available for only seven survey years, the poverty results are indicative only.

Conclusion: Taxation's welfare effects in Ghana are dimension-specific and mediated by government expenditure. Taxation contributes to delayed improvements in education and long-run improvements in healthcare access, but it does not significantly reduce income inequality. Poverty reduction appears to depend on how revenue is spent rather than on tax collection alone.

Keywords: Taxation, welfare, income inequality, education, healthcare, poverty reduction.

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List of Abbreviations

- **ADF** - Augmented Dickey-Fuller
- **AIC** - Akaike Information Criterion
- **AG** - Agriculture Value Added
- **ARDL** - Autoregressive Distributed Lag
- **CUSUM** - Cumulative Sum
- **DIT** - Direct Taxes
- **DITA** - Direct Taxes (% of GDP)
- **ECM** - Error Correction Model
- **ECT** - Error Correction Term
- **FCUBE** - Free Compulsory Universal Basic Education
- **FDI** - Foreign Direct Investment
- **GDP** - Gross Domestic Product
- **GDPPC** - Gross Domestic Product Per Capita
- **GEX** - Government Expenditure
- **GI** - Gini Index
- **GINI** - Gini Coefficient
- **GRA** - Ghana Revenue Authority
- **GRD** - Government Revenue Dataset (UNU-WIDER)
- **GSFP** - Ghana School Feeding Program
- **GSS** - Ghana Statistical Service
- **IDT** - Indirect Taxes
- **IDTA** - Indirect Taxes (% of GDP)
- **IMU** - Immunization Coverage
- **INF** - Inflation Rate
- **JB** - Jarque-Bera
- **LEAP** - Livelihood Empowerment Against Poverty
- **LM** - Lagrange Multiplier
- **NARDL** - Nonlinear Autoregressive Distributed Lag
- **NHIS** - National Health Insurance Scheme
- **NONTREV** - Non-Tax Revenue

- **NTRA** - Non-Tax Revenue (% of GDP)
- **OLS** - Ordinary Least Squares
- **OPN** - Economic Openness
- **PCGDPR** - Per Capita GDP Growth Rate
- **POVH** - Poverty Headcount
- **SD** - Standard Deviation
- **SHS** - Senior High School
- **SSER** - Secondary School Enrollment Rate
- **TDT** - Total Direct Tax Revenue
- **TIT** - Total Indirect Tax Revenue
- **TOT** - Total Tax Revenue
- **TOTA** - Total Tax Revenue (% of GDP)
- **TREV** - Total Tax Revenue
- **VAT** - Value-Added Tax
- **VECM** - Vector Error Correction Model
- **WDI** - World Development Indicators

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Dedication

I dedicate this work to God and my family.

Chapter 1 Introduction

1.1 Overview

This chapter provides the foundational background for the study, establishing the rationale and context from which the research problem emerges. It presents the statement of the problem, articulates the aims of the study, and justifies its necessity. Specific research objectives are outlined to guide the investigation, alongside research questions designed to address the identified problem. The chapter further examines the significance of the study to the Ghanaian context, delineates its scope and delimitations, and explains the organization of the thesis. The chapter concludes with a summary that encapsulates its key points and sets the stage for subsequent chapters.

1.2 Background to the Study

Ghana's economic landscape is characterized by traditional agriculture, emerging industries, and a growing services sector (Abbam, Johnson, Dash, & Padmadas, 2018). As one of the most stable and diversified economies in West Africa, Ghana has experienced steady economic growth in recent years (Jaurena, 2018). The country is endowed with rich natural resources such as gold, cocoa, and oil, which contribute significantly to export revenues (Niikoi Kotey et al., 2022). Agriculture remains a major source of employment for the population, while mining, construction, and other industries continue to expand (Eshun, 2018). The services sector, including finance, telecommunications, and tourism, has also experienced notable growth. Despite these

strengths, challenges such as income inequality, infrastructural deficits, and volatility in global commodity prices persist, highlighting the need to strengthen Ghana's economic resilience, sustainability, and inclusiveness.

Taxation serves as a critical instrument for government financing and socio-economic development (Kooshkebaghi, Emamgholipour, & Dargahi, 2022a). It is the primary source of revenue to maintain essential public services, including education, healthcare, social protection, defense, transportation, and public utilities. Beyond resource mobilization, taxation acts as a mechanism for income redistribution, helping to reduce disparities between high- and low-income groups. Effective tax collection enables the government to invest in human capital, improve overall economic conditions, and respond to unforeseen economic shocks, thereby underpinning the nation's socio-economic structure (Kooshkebaghi et al., 2022a; Franklin, Cintya, & Pablo, 2022).

Social welfare plays a pivotal role in promoting inclusive development and ensuring the well-being of citizens (Wokoma, 2023; Jensen & Thompson, 2020). Welfare programs—including social support, healthcare services, education, and other targeted interventions—enhance living standards, foster social cohesion, and reduce inequality (Gal & Bleikh, 2019). Accessible education and healthcare improve human capital, while welfare systems provide protection during periods of economic hardship, ultimately promoting a healthier, better-educated, and more productive population (Sahoo & Choudhury, 2023). Tax systems comprise multiple categories of levies, including personal income tax, value-added tax (VAT), and corporate tax (Andreyeva, Marple, Marinello, Moore, & Powell, 2022; Ayangbah & Sun, 2017). Income tax in Ghana is progressive, with rates ranging from 0% to 35% depending on income brackets. VAT, an indirect tax, is levied on the value added at each stage of production and distribution, with a standard rate of 15% (Ghana Revenue Authority, 2023). Corporate tax, imposed on firms' profits, is set at 25%. Additional levies include the Capital Gains Tax, Customs and Excise Duties, and the National Health Insurance Levy (Ghana Revenue Authority, 2023). Ghana's tax framework aligns with international standards while accommodating domestic economic priorities. Government reliance on taxation shapes fiscal policy and determines the ability to provide essential services (Ogundana, Ibidunni, & Adetoyinbo, 2017; A. B. Olarinde & Arike, 2022). Income tax, VAT, and corporate tax constitute the core of government revenue, while non-tax sources

such as fees and fines play a smaller role. Maintaining an effective and sustainable tax system requires systematic governance and careful policy implementation.

Ghana has implemented numerous social welfare programs to mitigate socio-economic hardships and promote inclusive development. The Livelihood Empowerment Against Poverty (LEAP) program provides cash transfers to vulnerable households, including older adults, persons with disabilities, and orphans (Kipo-Sunyehzi, 2021; Ardianti & Sari, 2023). The National Health Insurance Scheme (NHIS) aims to achieve universal health coverage by improving access to medical services, while the Free Senior High School (SHS) policy reduces financial barriers to education. The Ghana School Feeding Program (GSFP) improves nutrition and school attendance among children (Tette & Enos, 2020). Collectively, these programs address poverty, enhance education and health outcomes, and contribute to social development.

These initiatives demonstrate Ghana's commitment to leveraging taxation and welfare programs to support socio-economic development and reduce inequality. By examining the interplay between taxation and welfare, this study seeks to provide empirical insights to guide evidence-based policy reforms and strengthen the country's efforts toward inclusive growth.

1.3 Statement of the Problem

Taxes are a fundamental instrument for economic development in Ghana, providing the government with the resources necessary to fund public services and stimulate growth (Eshun, 2018; Kooshkebaghi et al., 2022a). Ghana relies heavily on taxation to support critical programs, including healthcare, education, and poverty alleviation initiatives (Kipo-Sunyehzi, 2021; Tette & Enos, 2020). However, concerns have emerged regarding the extent to which these taxes influence the welfare of Ghanaians, particularly among vulnerable populations (Anyaduba & Otulugbu, 2019; Omesi & Appah, 2021). Like many developing economies, Ghana faces multi-faceted challenges in aligning taxation with social welfare objectives. The efficiency and equity of tax policies are critical determinants of socio-economic outcomes, with indirect taxes, such as value-added tax (VAT) and excise duties, raising questions about their disproportionate burden on low-income households (Andoh & Nkrumah, 2022; Andreyeva et al., 2022). Such regressive

effects may exacerbate inequality and impede efforts to reduce poverty (Piketty & Saez, 2007; Palić, Hodžić, & Dumičić, 2019). The substantial informal sector in Ghana further complicates the taxation–welfare nexus. Operating largely outside the formal tax system, this sector presents significant challenges to revenue mobilization and equitable distribution of tax burdens (Aringo, 2023; Anim, Awotwe, Nyarku, & Kusi, 2020). These difficulties are compounded by limitations in tax compliance, enforcement, and institutional capacity, which can undermine the effectiveness of taxation in promoting social welfare (Ardianti & Sari, 2023; Cantero-Galiano, 2022). Given these considerations, there is an urgent need to investigate the relationship between taxation and welfare in Ghana. This study seeks to examine the structure, progressivity, and distributional effects of the tax system across different income groups, while also evaluating the compliance challenges faced by taxpayers and the differential impact of taxes on various segments of society. Furthermore, it assesses the efficiency and effectiveness of welfare programs financed through taxation, considering factors such as governance, resource allocation, and infrastructure constraints (Jensen & Thompson, 2020; Franklin et al., 2022).

To achieve these objectives, the research adopts a quantitative approach, analyzing tax data from the Ghana Revenue Authority, World Development Indicators, and the Ghana Statistical Service (Ghana Revenue Authority, 2023; Abbam et al., 2018). The analysis examines trends, progressivity, and distributional impacts, alongside socio-economic indicators including income distribution, GDP growth, and access to education and healthcare (Li & Bai, 2022; Safarova, 2021; Morrisson, 2002). Ultimately, this study aims to generate empirical evidence on the effects of taxation on welfare outcomes in Ghana. The findings are expected to provide actionable recommendations for policymakers, focusing on tax reform, improved compliance, and the optimization of welfare programs. By elucidating the relationship between taxation and social welfare, this research contributes to evidence-based policy formulation that promotes inclusive and sustainable development.

1.4 Purpose of the Study

This research aims to investigate how taxation policies influence social welfare in Ghana. Taxation is intrinsically linked to economic, social, and political structures; understanding its effects

on welfare provides a foundation for developing more equitable and effective fiscal systems. The study seeks to generate insights that can inform policymakers, economists, and social scientists on the welfare implications of different tax policies, both at the community and national levels.

Specifically, the study examines the relationship between taxation and welfare by analyzing how various types of taxes impact access to healthcare, education, income distribution, and poverty reduction. It evaluates the effectiveness of different tax systems in promoting poverty alleviation, economic mobility, and individual investment. The research also considers the trade-offs between taxation and economic efficiency, exploring the extent to which citizens are willing to contribute for enhanced social welfare.

Furthermore, the study assesses taxation's role in advancing social equity by examining its effects on access to and quality of essential public services, including healthcare, education, and social protection mechanisms. Particular attention is given to disadvantaged groups, such as low-income households and marginalized communities, to evaluate how tax policy changes affect the distribution of welfare across different segments of society.

Based on these findings, the research will provide evidence-based policy recommendations aimed at maximizing the positive impact of taxation on social welfare. The recommendations will be contextualized for local and national settings and will highlight avenues for future research, thereby contributing to the development of tax systems that better serve societal welfare needs.

1.5 Research Objectives

1.5.1 General Objective

To examine the impact of taxation on welfare, focusing on its effects on income inequality, access to education, access to healthcare, and poverty reduction in Ghana.

1.5.2 Specific Objectives

1. To assess the effect of taxes on income inequality in Ghana.
2. To determine the effect of taxes on access to education in Ghana.
3. To determine the effect of taxes on access to healthcare in Ghana.
4. To assess the effect of taxes on poverty reduction in Ghana.

1.6 Research Questions

1. What is the effect of taxes on income inequality in Ghana?
2. What is the effect of taxes on access to education in Ghana?
3. What is the effect of taxes on access to healthcare in Ghana?
4. What is the effect of taxes on poverty reduction in Ghana?

1.7 Significance of the Study

The study of the impact of taxation on welfare in Ghana is of considerable importance from both policy and research perspectives. Understanding the interdependencies between taxation and social welfare is essential for evaluating how effectively taxes serve their intended purposes and assessing the socio-economic consequences of changes in tax policy. This study provides a systematic examination of how taxation influences income distribution, access to public services, and poverty alleviation, offering evidence to inform sound policy decisions.

First, the findings of this research can serve as an essential reference for Ghanaian policymakers in the formulation and implementation of tax policies that align more closely with national welfare objectives. By identifying the strengths and weaknesses of the existing tax structure, policymakers can design a system that is both equitable and efficient. Such reforms may involve

strategic allocation of public expenditures to areas where they are most needed, including social protection, education, healthcare, and poverty reduction programs that directly benefit the most vulnerable populations.

Second, the study's focus on the role of taxation in poverty alleviation provides insights into the mechanisms underlying income distribution and socio-economic inequality in Ghana. By identifying aspects of the tax system that may be regressive, the research enables interventions to mitigate adverse effects on low-income households, thereby promoting more just and inclusive economic growth. Evidence from this study can guide policies aimed at improving the quality of life and reducing economic disparities across different segments of the population.

Furthermore, investigating the linkages between taxation and the provision of public services, such as education and healthcare, can clarify the extent to which tax revenues are effectively translated into tangible benefits for citizens. This analysis supports evidence-based decisions regarding resource allocation, ensuring that public funds are directed toward high-quality, accessible services that strengthen human capital and foster sustainable socio-economic development.

In addition to its domestic relevance, the study contributes to the broader discourse on taxation and welfare in developing economies. Researchers, economists, and international development agencies can use the findings as a reference for comparative studies and cross-national policy analysis, facilitating the adoption of best practices and evidence-informed approaches.

Overall, this research aims to provide valuable insights for policymakers, supporting the development of taxation policies that promote inclusive and sustainable development in Ghana. By elucidating the relationship between taxation and welfare, the study seeks to inform strategies that advance equity, social justice, and economic prosperity for all citizens.

1.8 Delimitation of the Study

The scope of this study is confined to examining the direct relationship between taxation and social welfare in Ghana. Specifically, the research investigates how taxation policies influence citizens' welfare, focusing on government-imposed taxes, including income tax, consumption

tax, property tax, and social security contributions.

The study emphasizes the impact of taxation on key welfare dimensions, such as access to basic public services, income distribution, social protection, and overall economic development. It does not attempt to evaluate broader macroeconomic policies or other factors that may indirectly affect welfare outcomes over the long term, such as the quality of education, healthcare system efficiency, or external trade dynamics.

By identifying the specific linkages between taxation and welfare, this research aims to provide evidence-based insights that can guide policymakers in designing a more equitable and effective tax system to enhance citizen well-being.

1.9 Organization of the Study

This thesis is structured into six chapters, each addressing distinct aspects of the study.

Chapter One introduces the research, providing the background, purpose, and objectives of the study. It formulates the research questions, discusses the significance of the study, outlines delimitations, defines key terms, and presents an overview of the thesis structure. The chapter concludes with a summary of its main points.

Chapter Two presents a comprehensive review of the literature relevant to the study. This chapter discusses major concepts, theoretical perspectives, and empirical studies related to taxation and welfare. It also outlines the formulation of a conceptual framework that guides the research. The chapter ends with a synthesis of the reviewed literature and a summary of key findings.

Chapter Three describes the research methodology. It introduces the study area, research design, and the rationale for the chosen approach. The chapter provides detailed descriptions of the study population, sampling techniques, and data collection methods. It also explains the analytical procedures, ethical considerations, and limitations of the study, concluding with a summary of the methodological approach.

Chapter Four reports the research findings. It presents an introduction to the results, followed by detailed analyses of the collected data, response rates, and outcomes, referencing relevant tables and figures. The chapter concludes with a concise summary of the findings.

Chapter Five is dedicated to the discussion of the results. It interprets the findings in relation to the study objectives and contextualizes them within existing literature. Key discoveries are highlighted and critically analyzed to understand their implications for taxation and welfare in Ghana. The chapter ends with a summary of the discussion.

Chapter Six concludes the thesis by presenting the overall findings, drawing conclusions, and offering evidence-based recommendations. This chapter summarizes the study's contributions, policy implications, and potential directions for future research.

1.10 Chapter Summary

Chapter One of this thesis establishes the foundation for the study by providing a comprehensive background on the role of taxation as a fiscal policy instrument and its impact on social welfare. The chapter traces the evolution of taxation in Ghana, from traditional African societies through the colonial period and post-independence challenges, highlighting key transitions and reforms.

The problem statement clearly defines the primary objective of the study: to examine the relationship between taxation and social welfare by comparing the effects of progressive and regressive tax systems in Ghana and evaluating whether taxation policies can effectively promote social welfare. The chapter also articulates the research purpose, emphasizing the relevance of both policy formulation and scholarly investigation regarding taxation and welfare outcomes.

Research objectives and questions are outlined to guide the study, focusing on critical aspects such as income distribution, poverty alleviation, and potential reforms of the tax system. Furthermore, the significance of the study is discussed, illustrating its potential contributions to policymakers, resource allocation, and future research on taxation and welfare systems.

The delimitation statement specifies the scope of the study, clarifying that the focus is on taxation and welfare in Ghana, rather than broader social outcomes such as life expectancy

or general health indicators. Finally, the chapter provides an overview of the thesis structure, outlining the content and purpose of each subsequent chapter to guide the reader through the research process.

Chapter 2 Review of Literature

2.1 Introduction

This study focuses on the interplay between taxation and welfare in Ghana. This chapter provides a conceptual framework for understanding key ideas in the field and critically appraises relevant research. It presents a comprehensive review of the literature, highlighting the issues most pertinent to the study's research focus. All relevant findings, empirical studies, and reports from prior research have been carefully examined and analyzed to inform the study and establish its scholarly context.

2.2 Concept Review

This subsection reviews key concepts related to taxation and welfare within the context of the study area. For taxation, the review focuses on fundamental ideas such as the definition of taxation, direct and indirect taxes, income taxes, and consumption taxes. Regarding welfare, the discussion encompasses concepts such as welfare itself, income inequality, education, healthcare, and poverty reduction. The purpose of this review is to provide a conceptual foundation that guides the analysis of the relationship between taxation policies and social welfare in Ghana.

2.2.1 Concept of Taxation

Taxation refers to the revenue collected by state governments from individuals and businesses (Omesi & Appah, 2021). It serves as a mechanism for raising funds to compensate for externalities, stabilize the economy, and ensure equitable income distribution(Owuru & Olabisi, 2020). Anyaduba and Otulugbu (2019) define taxation as the collection of funds from people, enterprises, or other entities to finance public services and government expenditures, including income tax, sales tax, and property or corporate taxes. Similarly, (Palić et al., 2019) describe taxation as the government's levying of taxes on individuals and corporations, such as personal income tax, corporate profits, or declared turnover, at legally determined rates (Tadesse, 2023). The primary purpose of taxation is to expand public revenues and generate resources for government programs such as infrastructure development, environmental protection, healthcare, education, social welfare, and defense (Sahari et al., 2020; Kooshkebaghi, Emamgholipour, & Dargahi, 2022b). Taxation also functions as an instrument of wealth redistribution and economic stability. Progressive tax systems, in which higher earners pay a larger proportion of their income, enable the redistribution of wealth to lower-income groups (Ogundana et al., 2017; Ceyhan, Köstekçi, & Göv, 2021).

Moreover, taxation influences economic behavior and consumer preferences. Governments can use tax incentives or deductions to encourage specific activities, such as investment in renewable energy or research and development (Pokharel, 2018; Ntiamoah & Asare, 2020). However, while taxation is essential for financing public goods and services, it can also generate public debate regarding fairness, efficiency, and its impact on economic development (Fikr, Johari, Sholeh, & Suprayitno, 2021; Nemec, Pompura, & Šagát, 2015). Striking a balance between generating revenue and minimizing costs for taxpayers remains a persistent challenge for governments worldwide (Cantero-Galiano, 2022). In summary, taxation is a critical tool for revenue generation, social equity, and economic policy. Nevertheless, it continues to present challenges related to fairness, efficiency, and public acceptance, making it an enduring focus of research and policy deliberation.

2.2.2 Concept of Direct Taxation

Direct taxation is a fundamental component of fiscal policy and involves the imposition of taxes directly on individuals or entities by a government authority (Rahmat, Abd Hamid, Razak, & Sulaiman, 2023). Unlike indirect taxes, which are levied on goods and services and passed on to consumers, direct taxes are imposed directly on the income, wealth, or property of taxpayers (Kryeziu, 2021). Over time, the concept of direct taxation has evolved in response to changing economic structures and political ideologies (Tikhonova, Telegina, Babanskaya, & Grudneva, 2019). Historically, direct taxation primarily consisted of taxes on land, property, and income (Balamatsias, 2018). However, with economic modernization and diversification, the scope of direct taxation expanded to include taxes on personal income, corporate profits, capital gains, and inheritance. This evolution reflects governments' need to broaden revenue sources to finance public expenditures in increasingly complex economies.

Direct taxation is characterized by several key features. First, it promotes transparency because taxpayers are aware of the amount they pay and the basis upon which it is calculated (Yilmaz, Ozyer, & Ozyer, 2019). Second, many direct taxes are progressive, meaning that the tax rate increases as income or wealth increases. This progressivity helps ensure a more equitable distribution of the tax burden across income groups. Third, direct taxation serves as an important instrument for income redistribution and the reduction of socioeconomic inequalities. Scholars broadly agree that the defining feature of direct taxation is its direct relationship between the taxpayer and the government without intermediary transactions(Khoruzhy et al., 2020; Das Gupta & Gupta, 2017). This distinguishes direct taxes from indirect taxes, which are usually collected through intermediaries such as businesses.

In Ghana, direct taxation is governed by the Income Tax Act, 2015 (Act 896), which provides the legal framework for assessing and collecting taxes on individual and corporate income (Anim et al., 2020). The Act covers several forms of direct taxes, including personal income tax, corporate income tax, and capital gains tax. The Ghana Revenue Authority (GRA) is responsible for the administration and enforcement of these tax laws (Anim et al., 2020). In summary, direct taxation constitutes an essential element of public finance, involving the direct imposition of taxes on individuals and organizations based on income, wealth, or property. It is characterized

by transparency, progressivity, and a direct link between the taxpayer and the government.

2.2.3 Concept of Indirect Taxation

Indirect taxation refers to a form of taxation in which the burden of the tax is shifted from the entity that initially pays the tax to another party, typically the final consumer (Yilmaz et al., 2019). This occurs because indirect taxes are imposed on goods and services at the point of production or consumption rather than directly on individuals or businesses (Fredrick & Peter, 2019). Historically, indirect taxes have included levies such as excise duties, value-added taxes (VAT), and customs duties. These taxes are usually collected by intermediaries such as manufacturers, wholesalers, or retailers, who subsequently incorporate the tax into the price of goods and services. As a result, the final consumer ultimately bears the tax burden.

Over time, the scope of indirect taxation has expanded to include additional forms of regulatory taxes such as environmental taxes and so-called "sin taxes," which are imposed on products such as tobacco and alcohol. This evolution reflects the increasing complexity of modern economies and governments' efforts to raise revenue while simultaneously influencing consumer behavior. One of the main characteristics of indirect taxation is its regressive nature, as it often places a relatively higher burden on lower-income households compared to wealthier individuals. This occurs because low-income households typically spend a larger proportion of their income on consumption, which is subject to indirect taxation.

Researchers widely acknowledge that indirect taxes serve dual roles: generating government revenue and influencing market behavior (Lu, Zhao, Li, & Cao, 2023). By adjusting tax rates on specific products, governments can encourage or discourage particular types of consumption, thereby supporting broader policy objectives such as public health protection, environmental sustainability, or economic development.

In Ghana, indirect taxation is governed by several statutory frameworks. The Value Added Tax Act, 2013 (Act 870) regulates the administration of VAT, while the Excise Duty Act, 2014 (Act 878) provides for the imposition of excise duties on specific goods. Additionally, the Customs Act, 2015 (Act 891) governs the collection of customs duties on imported goods.

2.2.4 Concept of Income Tax

Income tax refers to a tax imposed on income earned by individuals and corporations within a specific tax period (Chumaidi & Noor, 2022). It represents one of the most significant sources of government revenue and is widely used to finance public expenditures and essential government services. Income taxation plays a critical role in modern fiscal systems by ensuring that individuals and corporations contribute a portion of their earnings to support national development. Governments rely on income taxes to fund public goods and services such as infrastructure, healthcare, education, and social protection programs. In addition to revenue generation, income taxation also contributes to economic stability and social equity by redistributing wealth and reducing income disparities.

Under most tax systems, individuals and businesses are required to declare their income and calculate the tax payable according to specified tax rates and regulations. These systems often incorporate progressive tax structures, in which higher income levels are taxed at higher rates. Such systems are designed to ensure fairness and enhance income redistribution within society. Furthermore, income taxation can influence economic behavior through tax incentives and deductions. Governments often introduce tax credits or exemptions to encourage investments in sectors such as renewable energy, research and development, and education. These incentives help stimulate economic growth while aligning individual economic activities with national development goals. Overall, income tax is a central instrument of fiscal policy, serving multiple purposes including revenue mobilization, redistribution of income, and the promotion of economic development.

2.2.5 Concept of Personal Income Tax

Personal income tax refers to a tax levied on income earned by individuals from various sources such as wages, salaries, interest, dividends, and rental income (Omesi & Appah, 2021). It is a form of direct taxation imposed by governments on individuals within their jurisdiction (Kato & Toyofuku, 2018). Personal income tax constitutes an important source of government revenue and plays a crucial role in financing public services and social programs. Historically,

income taxes were primarily imposed on wealthy individuals; however, modern tax systems apply personal income tax across multiple income brackets, thereby broadening the tax base.

Over time, the definition of personal income tax has expanded to include various forms of income beyond wages and salaries. These now include dividends, interest income, rental earnings, capital gains, and income from self-employment. Modern tax systems also incorporate deductions and tax credits to encourage socially desirable behaviors such as homeownership, retirement savings, and educational investment. One of the most important characteristics of personal income tax is its progressive structure, in which tax rates increase with rising income levels (Gobachew, Debela, & Shibiru, 2018). This progressive design helps promote equity in the tax system and contributes to reducing income inequality. Personal income tax therefore serves not only as a revenue-generating mechanism but also as a policy tool for addressing income disparities and promoting economic justice.

2.2.6 Gini Coefficient

The Gini coefficient is a widely used measure of inequality in income distribution, calculated based on the Lorenz curve (Omesi & Appah, 2021). The Lorenz curve, developed by Lorenz (1905), graphically represents the distribution of income or wealth in a society, showing the cumulative proportion of income held by cumulative population segments (Didenko, Skripnuk, & Kikkas, 2017; Fellman, 2018). The Gini coefficient quantifies deviation from perfect equality, with values ranging between 0 and 1: 0 indicates absolute equality, whereas values closer to 1 indicate higher income inequality (Hasell, 2023).

Income inequality can occur within a country or between countries. For instance, Ortiz and Cummins (2011) reported that in 2008, the average Gini coefficient in Sub-Saharan Africa was 0.442, compared to 0.483 for Latin America, 0.354 for Eastern Europe and Central Asia, and 0.309 for high-income countries. Other measures, such as decile or quintile ratios, are also used to evaluate income distribution. Nevertheless, the Gini coefficient remains the most popular and widely applied metric in empirical studies on income inequality (Hasell, 2023; Anyaduba & Otulugbu, 2019).

2.2.7 Concept of Access to Education

Access to education is a fundamental right and a cornerstone of social and economic development. It encompasses the availability, affordability, and inclusivity of educational opportunities for all individuals, irrespective of socioeconomic background, gender, or disability status. Access enables individuals to acquire knowledge, develop skills, and reach their full potential, thereby promoting social mobility and reducing inequality (UNESCO, 2021).

In Ghana, access to education is constitutionally recognized and supported by government policies. Initiatives such as the Free Senior High School (SHS) program have reduced financial barriers, enabling more students to enroll in secondary education (Government of Ghana, 2017). Programs addressing gender disparities, including the Ghana School Feeding Program and the Capitation Grant, have improved attendance and retention rates, particularly among girls from marginalized communities (Ministry of Education, 2019). Additionally, the Inclusive Education Policy seeks to accommodate children with disabilities within mainstream schools, ensuring equitable learning opportunities (Ministry of Education, 2019).

Despite these efforts, challenges persist, including inadequate infrastructure, limited educational resources, and regional disparities. Continuous investment and policy reforms are required to ensure that access to education in Ghana is inclusive, equitable, and of high quality.

2.2.8 Concept of Access to Healthcare

Access to healthcare refers to the ability of individuals and communities to obtain necessary medical services, including preventive, curative, and rehabilitative care, in a timely and affordable manner (World Health Organization, 2018). It involves factors such as physical proximity to healthcare facilities, financial resources, health insurance coverage, and cultural or language considerations (Andersen & Davidson, 2017; Penchansky & Thomas, 1981). Equitable access also depends on the quality, acceptability, and equity of healthcare provision.

In Ghana, the National Health Insurance Scheme (NHIS), implemented in 2003, represents a major policy intervention to reduce financial barriers to healthcare, particularly for marginalized

populations (Agyepong et al., 2016). Nevertheless, disparities persist, especially between urban and rural areas, due to challenges in transportation, infrastructure, and availability of skilled healthcare professionals (Gal & Bleikh, 2019). Cultural practices and beliefs also influence healthcare utilization, underscoring the importance of culturally sensitive interventions.

Quality differences further complicate healthcare access, with urban centers often offering more advanced facilities and specialized personnel compared to rural regions (Yawson, Aikins, & Antwi, 2019). Continued investment in healthcare infrastructure, workforce development, and policy implementation is essential to ensure equitable access across Ghana.

2.2.9 Concept of Poverty Reduction

Poverty reduction encompasses strategies aimed at alleviating and ultimately eradicating poverty within specific populations or geographic areas (Mbughi, 2018). It involves multidimensional interventions targeting income inequality, access to basic services, employment, and sustainable economic activities (World Bank, 2020; Ravallion, 2016). Effective poverty reduction strategies include income support programs, social protection mechanisms, access to education and healthcare, and community empowerment initiatives (United Nations Development Programme, 2018; D. Narayan, Chambers, Shah, & Petesch, 2000).

In Ghana, poverty reduction is a key policy priority, addressed through programs such as the Livelihood Empowerment Against Poverty (LEAP) program and the Ghana School Feeding Program (Ghana Ministry of Gender, Children and Social Protection, 2020; Ghana School Feeding Programme, 2020). Education policies, including the Free SHS initiative, aim to break intergenerational cycles of poverty by providing equitable learning opportunities for low-income households (Government of Ghana, 2017). Expansion of the NHIS and healthcare infrastructure further supports vulnerable populations (National Health Insurance Authority, 2020). Economic empowerment initiatives targeting agriculture, microfinance, and skills development also enable individuals and communities to generate income and achieve self-reliance (Ministry of Food and Agriculture, 2017). Nonetheless, regional disparities in access to education, healthcare, and economic opportunities remain significant (Ghana Statistical Service, 2018).

In conclusion, Ghana's poverty reduction strategy combines social safety nets, education and healthcare interventions, and sustainable economic empowerment programs. While progress has been achieved, continuous efforts are needed to address inequalities and enhance the effectiveness of these strategies.

2.3 Theoretical Review

This chapter presents the theoretical framework underpinning this study, with a focus on the multifaceted relationship between taxation and socio-economic outcomes, including inequality, education, healthcare, and poverty reduction. The theoretical discussion integrates classical and contemporary perspectives, encompassing the Endogenous Growth Model, Ability-to-Pay Theory, Human Capital Theory, Theory of Infrastructure-led Development, Benefit Theory, Theory of Realization and Exigent Implementation, and the social welfare concept. Together, these frameworks provide a comprehensive lens for examining how taxation affects public resource allocation, individual well-being, and societal development. The selected theories were chosen based on their explanatory power in addressing the distributive, economic, and developmental implications of taxation policies in contemporary contexts.

2.3.1 The Benefit Theory

The Benefit Theory of taxation proposes that taxpayers should contribute to government revenues proportionally to the benefits they receive from public services (Ayodeji & Others, 2021). This theory emphasizes an equity-efficiency trade-off, suggesting that the allocation of tax burdens can reflect both usage and derived advantages. For instance, those who gain substantial benefit from public infrastructure, healthcare, or security services are expected to contribute commensurately through taxation. This approach aligns with market-like mechanisms in public finance, where taxes serve as a price signal for public goods and help guide the government in resource allocation (Hermawan & Sinaga, 2020). Morimura (2019) conceptualizes proportional income tax as a form of benefit taxation, asserting that individuals' tax contributions should mirror their economic advantages from state protection. Peters further highlights that the so-

cial, legal, and economic frameworks established by governments facilitate income generation and security, creating a proportional link between individual benefits and tax obligations. Thus, the Benefit Theory supports policies that balance equity and efficiency while targeting income redistribution indirectly.

The practical application of the Benefit Theory is illustrated through excise taxes, such as the gasoline tax, where revenues are earmarked for public goods like road construction and maintenance. Consumers with higher consumption levels contribute more, reflecting greater derived benefits. Despite its theoretical appeal, the Benefit Theory encounters limitations in practice. Most public goods, including education and healthcare, are consumed collectively, making precise benefit measurement challenging. Furthermore, strict adherence may disproportionately impact low-income groups who, relative to their income, derive significant utility from public services. Consequently, modern tax policy often integrates the Benefit Theory with progressive taxation principles to achieve both efficiency and equity. In summary, the Benefit Theory provides a normative framework for linking taxation to service provision, illustrating how government revenue collection can reflect societal preferences while mitigating inequitable distributions.

2.3.2 Ability-to-Pay Theory

The Ability-to-Pay Theory posits that individuals should be taxed according to their financial capacity, typically measured by income, wealth, or consumption (Jhingan, 2014). This theory is grounded in principles of fairness and justice, advocating progressive taxation where higher earners contribute a larger share of public revenue. Its significance lies in the direct potential to reduce income inequality by redistributing resources to fund social services, infrastructure, and welfare programs. Tax systems based on the Ability-to-Pay principle also support fiscal sustainability by ensuring that those most able to contribute bear an appropriate share of the revenue burden. The theoretical rationale extends beyond income distribution; it underscores societal obligations to fund collective goods and services, such as public education, healthcare, and social protection programs, which benefit all citizens, regardless of whether they directly utilize these services.

Application of the Ability-to-Pay Theory in addressing income inequality involves progressive taxation schemes that balance fiscal responsibility with social justice. Critics, however, highlight limitations. Jhingan (2014) argues that reliance solely on consumption expenditure ignores wealth accumulation and investment, potentially misrepresenting taxpayers' true capacity. Appah (2019) notes that excessively progressive systems may discourage productivity and savings, while Omesi and Appah (2021) affirm that appropriately structured policies can promote welfare maximization without significant economic distortion. In practice, governments often combine this theory with other tax frameworks, including indirect taxes and benefit-based levies, to ensure both fairness and revenue efficiency. Overall, the Ability-to-Pay Theory remains central to contemporary tax policy design, emphasizing the link between income, capacity, and societal obligations, while serving as a mechanism for reducing economic disparities.

2.3.3 Human Capital Theory

Human Capital Theory emphasizes the economic value of investments in knowledge, skills, and health, asserting that such investments enhance individual productivity and societal prosperity (Zimmer, 2022; Jiang & Shi, 2023). Human capital encompasses formal education, vocational training, and health interventions, all of which collectively increase labor market productivity, wage potential, and innovation capacity (Yaqoob & Others, 2022). The theory asserts that governments and individuals should view expenditure on education and healthcare as investments with quantifiable economic returns. Policies that expand access to schooling, vocational programs, and healthcare services directly contribute to human capital accumulation, facilitating long-term economic growth and social mobility. Human Capital Theory also accounts for lifelong learning, emphasizing continuous skill development as essential for adapting to technological advances and labor market shifts (Rothomi & Rafid, 2023). Health, as a component of human capital, significantly influences productivity and economic participation; healthier populations are more capable of sustaining economic activities, contributing to social welfare (Kucharska, 2022).

Application of Human Capital Theory in taxation and education highlights the role of public investment in human development. Progressive taxation, informed by the Ability-to-Pay

principle, funds educational infrastructure, scholarships, and training programs, ensuring equitable access to opportunities (A. Olarinde & Arike, 2022). By financing human capital, governments enhance labor productivity, reduce income disparities, and stimulate sustainable economic growth. Critiques of the theory note its potential oversimplification of socio-economic inequalities and the complexity of skill acquisition (MacKenzie & Others, 2023; Yaqoob & Others, 2022). Nevertheless, Human Capital Theory remains foundational for understanding the role of taxation in promoting education, health, and workforce development, demonstrating the interdependence of fiscal policy and human development outcomes.

2.3.4 Theory of Infrastructure-led Development

The Theory of Infrastructure-led Development posits that strategic investments in foundational infrastructure are essential drivers of economic growth and societal transformation (Buthelezi, 2021). Infrastructure facilitates the efficient movement of goods, people, and information, enhancing productivity and enabling access to services such as education, healthcare, and commerce. Well-planned infrastructure reduces transaction costs, promotes business development, and catalyzes technological adoption. Within the context of taxation, revenue allocation to infrastructure development ensures that public resources are used to support productive investments that yield long-term societal benefits.

Application of this theory to taxation and education suggests that investments in school buildings, educational technology, and connectivity improve learning outcomes and human capital productivity. Infrastructure investments complement educational initiatives by creating environments conducive to learning and enhancing equitable access across socio-economic groups. Tax-funded infrastructure, aligned with strategic development goals, therefore becomes a pivotal mechanism for fostering economic growth, reducing inequality, and supporting sustainable human capital accumulation (A. Olarinde & Arike, 2022). By combining infrastructure-led development with progressive taxation, governments can simultaneously promote equity, efficiency, and long-term economic resilience.

2.3.5 Theory of Realization and Exigent Implementation

The Theory of Realization and Exigent Implementation asserts that taxation should yield tangible, timely, and beneficial outcomes for taxpayers without causing undue economic strain (O. Adegbite, 2021). This theory emphasizes the accountability of government in deploying revenues to deliver essential services, including healthcare, education, and public goods. Tax systems guided by this framework ensure that citizens perceive a direct linkage between contributions and public benefits, reinforcing compliance and trust.

In healthcare, this theory underscores that tax revenues should directly improve access, quality, and equity of medical services, particularly for marginalized populations. Ineffective utilization of tax revenue diminishes public confidence and undermines the potential of fiscal policies to generate social welfare. By ensuring the prompt and equitable allocation of resources, this theory bridges the gap between fiscal policy design and societal outcomes, highlighting the ethical and practical dimensions of taxation.

2.3.6 Social Welfare and Taxation for Poverty Reduction

The concept of social welfare provides a normative foundation for assessing taxation in relation to poverty reduction (Maniquet & Neumann, 2021). Maniquet and Neumann (2021) propose tax schemes that balance individual well-being with poverty alleviation, emphasizing that labor income taxation should minimize the effort required to achieve basic living standards. By integrating considerations of leisure, productivity, and subsistence, this framework aligns fiscal policy with social welfare objectives. Taxation under this model supports equitable redistribution, funding social protection, healthcare, education, and poverty reduction programs. While practical limitations exist, including labor market frictions and imperfect information, the social welfare framework remains a crucial guide for designing tax policies that enhance societal well-being and reduce economic inequality.

2.4 Empirical Review

2.4.1 Taxation and Income Inequality

Several empirical studies have investigated the relationship between taxation and income inequality, particularly in the context of developing economies like Nigeria. Omesi and Appah (2021) analyzed the impact of different tax types, including personal income tax, company income tax, petroleum profit tax, customs duties, excise duties, and value-added tax on the Gini coefficient as a measure of income inequality. Using empirical analysis and literature review, the study revealed that personal and company income taxes exhibit a significant negative relationship with income inequality, indicating that progressive taxation can contribute to reducing disparities. Conversely, petroleum profit tax was positively associated with income inequality, whereas value-added tax showed an insignificant positive correlation. The study highlighted that mismanagement of government spending on health and education can exacerbate inequality, emphasizing the need for effective allocation of revenue and enhanced tax compliance. Despite offering valuable insights, the study faced limitations, such as restricted geographic focus, limited discussion of potential data biases, and omission of certain socio-economic factors that may influence inequality.

Building on this, Xue (2018) conducted a literature-based analysis to assess the redistributive impact of direct taxes in Nigeria. Their findings indicated a strong positive relationship between revenue from personal income tax and petroleum profit tax and improved income redistribution, while company and education taxes had minimal impact. The study argued that taxation, particularly direct taxation, not only serves as a revenue-generating mechanism but also plays a central role in achieving equitable income distribution, thereby addressing structural imbalances. The authors emphasized the dual importance of robust tax administration and strategic allocation of resources to facilitate social welfare and infrastructure development.

In contrast, Obaretin et al. (2017) utilized a 34-year time series dataset to examine the influence of taxation on the Gini coefficient in Nigeria. The results suggested that direct, indirect, and total taxes did not significantly affect income inequality, indicating that the Nigerian tax sys-

tem may not adequately function as a redistributive tool. While the study employed rigorous statistical methods, including trend analysis, its reliance on secondary data, absence of detailed evaluation of individual tax types, and focus solely on Nigeria limited the generalizability of the findings.

Collectively, these studies demonstrate a nuanced and context-dependent relationship between taxation and income inequality. Progressive direct taxes, when effectively collected and combined with strategic allocation of public resources to education and healthcare, can significantly reduce income disparities. However, indirect taxes or poorly administered tax structures may fail to achieve this outcome, highlighting the importance of institutional capacity, governance quality, and complementary fiscal policies. Future research should explore cross-country comparisons, consider the impact of different tax instruments in varying institutional contexts, and incorporate comprehensive socio-economic variables to generate robust, policy-relevant insights.

2.4.2 Taxation and Education

Empirical research examining the relationship between taxation and education financing highlights the pivotal role of government revenue in supporting educational development. A. Olarinde and Arike (2022) investigated the impact of taxation on education financing in Nigeria using econometric approaches, including regression models, cointegration analysis, Vector Error Correction Models (VECM), and Granger causality Wald tests. Their study revealed a positive and statistically significant relationship between taxation and educational financing, demonstrating that increased tax revenue allows governments to improve education services, expand access, and enhance the quality of education. These findings corroborate earlier research by Babalola and Aminu (2011), who similarly highlighted taxation as a critical mechanism for mobilizing resources for educational development in resource-constrained settings.

Oluyebo (2022) analyzed the influence of petroleum profit tax and education tax on education outcomes and broader economic development in Nigeria. The study found a positive correlation between petroleum profit tax revenue and tertiary school enrollment, as well as life expectancy at birth, indicating that taxation can indirectly support both human capital formation and health outcomes. Interestingly, education tax revenue displayed a negative relationship with tertiary

enrollment, which may reflect inefficiencies in collection or allocation mechanisms. The study also observed that both taxation streams positively influenced per capita gross national income, highlighting the broader socio-economic impact of effective tax mobilization on educational and economic outcomes.

Ayangbah and Sun (2017) investigated the effect of taxation on higher education affordability in Nigeria, concluding that government grants should be increased to cover rising tertiary education costs. While the study offered practical policy recommendations, its reliance on qualitative assessment and absence of quantitative data limited the robustness and generalizability of its findings. Additionally, the exclusive focus on Nigeria restricts cross-context applicability, especially for countries with different fiscal structures, such as Ghana.

Taken together, these empirical studies underscore the crucial nexus between taxation and education financing. Effective tax administration, combined with progressive and well-targeted tax policies, can enhance government capacity to fund education, thereby fostering human capital development. However, challenges persist, including inefficiencies in tax collection, misallocation of funds, and limited cross-country comparative evidence. Future research should examine the differential effects of direct versus indirect taxes on educational outcomes, evaluate allocation efficiency, and explore the interplay between taxation, governance, and education policy to provide actionable insights for policymakers aiming to strengthen human capital through fiscal interventions.

2.4.3 Empirical Review on Taxation and Healthcare

The relationship between taxation and healthcare provision has received considerable attention in empirical research, particularly in developing economies where public health systems are heavily dependent on government revenues. (A. Adegbite, 2021) conducted a comprehensive study examining the relationship between taxation and health services in Nigeria, employing advanced econometric techniques to analyze both short-run and long-run dynamics. The study utilized cointegration analysis to explore the long-run equilibrium relationship between taxation and health services and employed a Vector Error Correction Model (VECM) to capture the short-run adjustments. Findings revealed a significant positive relationship between taxation

and healthcare provision, suggesting that higher tax revenues facilitate increased government expenditure on healthcare services. The study also confirmed earlier findings by Adisa-Atta (2017) in Ghana, reinforcing the notion that taxation can serve as a crucial instrument for improving public health systems. Notably, Adegbite emphasized the importance of a sustained tax base and effective revenue mobilization as critical drivers for healthcare financing. While the study is methodologically robust, it is limited by its focus on Nigeria, and the generalizability of its results to other contexts remains uncertain.

In Ghana, Adisa-Atta (2017) examined public attitudes toward higher taxation or user fees intended to improve government healthcare expenditure. Using cross-tabulation, correlation analysis, and multiple linear regression, the study explored whether willingness to pay higher taxes correlated with demographic factors, healthcare access, perceptions of government performance, and perceived corruption. The study found that only 35% of respondents supported higher taxes or user fees, despite widespread challenges in accessing quality healthcare. Regression results indicated a positive association between support for higher taxation and perceived improvement in healthcare delivery, highlighting the potential role of citizen acceptance in effective health financing. The study's strength lies in its use of rigorous quantitative techniques to identify determinants of public willingness to fund healthcare, though it did not assess the longitudinal impact of tax increases on actual health service delivery.

Boyd (2014) investigated the potential impact of alternative healthcare spending scenarios on state and local government budgets in the United States, highlighting the substantial growth of healthcare expenditures over the past 25 years. The study reported that state and local government healthcare spending nearly doubled relative to the overall economy, driven by increases in Medicaid and non-Medicaid expenditures. The study is notable for its extensive analysis of budgetary trends, providing a macro-level perspective on healthcare financing. However, it did not examine the specific contribution of tax revenue to this growth, leaving unanswered questions regarding the direct link between taxation and health expenditure.

Doherty (2014) explored the relationship between tax revenue and public healthcare financing in South Africa, combining quantitative and qualitative analyses. The study demonstrated that increases in tax revenue do not automatically translate into higher government health expenditure; instead, broader fiscal and economic policies, such as tax rate adjustments and

debt repayment strategies, play a significant role. This study contributes to understanding the complex policy environment influencing health financing. However, it lacks a detailed year-by-year analysis of how tax allocations impact healthcare service delivery, limiting the ability to draw conclusions about causal relationships.

In summary, empirical evidence indicates a generally positive association between taxation and healthcare provision, though outcomes are highly context-dependent. While studies in Nigeria and Ghana emphasize the potential of taxation to improve public health services, research in South Africa and the United States highlights the influence of broader fiscal policy decisions. These findings underscore the importance of not only raising tax revenue but also ensuring effective allocation and management of funds to maximize health service delivery. Collectively, these studies provide a foundation for understanding taxation as a policy tool for healthcare financing while revealing critical gaps for further empirical investigation, including longitudinal assessments and cross-country comparisons.

2.4.4 Empirical Review on Taxation and Poverty Reduction

Adu, Alagidede, Osei, and Asamoah (2023) conducted a study on the relationship between tax systems, poverty, and inequality. The authors first assume a nonlinear effect of tax systems on poverty and inequality, which is largely absent in the existing literature. They move beyond standard nonlinear treatments by testing whether the relationships are asymmetric. Additionally, the study considers both income and non-income measures of poverty to provide a more comprehensive assessment. Taxes were decomposed into direct and indirect categories, and the total tax effect was analyzed. Time-series data from 1983 to 2016 were gathered from multiple sources. The authors applied the novel NARDL (Nonlinear Autoregressive Distributed Lag) estimation approach while using ARDL as a baseline technique to examine the asymmetric effects of tax systems on poverty and inequality. Results reveal that tax systems exert asymmetric effects, with indirect taxes exerting a more significant impact on poverty and inequality than direct taxes. The study recommends that policymakers improve the progressivity of indirect taxes and reduce reliance on them to enhance the distributional impact of Ghana's tax system.

Lofgren and Cicowiez (2022) conducted a study titled "Taxation, infrastructure investment,

growth, and poverty reduction: A case study of Zimbabwe," providing a detailed analysis of the interplay between taxation policies, infrastructure investment, economic growth, and poverty reduction in Zimbabwe. Using macro- and meso-level Social Accounting Matrices (SAMs) and cross-country data from the World Bank, the authors examined Zimbabwe's economic performance from 2000 to 2014. They applied a Computable General Equilibrium (CGE) model to simulate the impact of different policy scenarios on key macroeconomic indicators, including GDP, investment, and poverty rates. Their findings indicate that increased infrastructure investment, particularly in energy and transport, could substantially enhance economic growth and reduce poverty. The study further emphasizes that reforms in taxation, such as broadening the tax base and reducing corporate income tax rates, could generate additional resources for infrastructure investment and social spending. These insights highlight the complex interdependencies among taxation, public investment, and poverty alleviation.

Lustig (2017) provides an extensive analysis of fiscal policy effects on inequality and poverty in 29 low- and middle-income countries. The study employs fiscal incidence methodologies to analyze the distributional consequences of government revenues and spending, including personal income taxes, payroll taxes, direct transfers, and consumption taxes. Household surveys, using income or consumption as welfare indicators, were applied. Findings suggest that fiscal policy can effectively reduce poverty and inequality; however, the impact depends on the size and composition of government revenues and expenditures. The author stresses the importance of investing in education and healthcare as mechanisms for inclusive growth and poverty reduction. These results offer critical insights into the capacity of fiscal policy to enhance equitable development outcomes.

Hirvonen et al. (2016) examined the effectiveness of taxation and social protection in Ethiopia. Key findings include the success of the Productive Safety Net Program (PSNP) in reducing poverty, with potential for expansion to currently excluded areas. The study also highlights the risk of income taxation becoming regressive if tax brackets are not regularly adjusted for inflation. Petersen (2008) assessed decentralization, integration, taxation, and revenue-sharing strategies, concluding that decentralization can enhance efficiency, transparency, and social welfare, although improper implementation may exacerbate inequalities. Leventi et al. (2017) studied tax-benefit policies in seven European countries, finding that child benefits, housing sup-

port, and social assistance reduce poverty, but effectiveness varies by country context. Kanbur et al. (2018) analyzed optimal taxation and public provision in developing countries, emphasizing linear tax policies, the role of informality, and the impact of public spending on well-being and poverty reduction.

Overall, these studies collectively underscore the multifaceted and context-dependent relationship between taxation and poverty reduction. While taxation can serve as a significant tool for promoting social welfare, outcomes are contingent upon tax structure, policy progressivity, and efficient allocation of public resources. The literature reveals a critical need for further empirical studies across different countries and longitudinal analyses to comprehensively understand taxation's role in poverty alleviation.

2.5 Research Gaps

A research gap represents an area within a field of study that has been insufficiently explored or inadequately addressed in existing literature. Identifying such gaps is critical, as they delineate the problem that the current research aims to address. Although extensive scholarly inquiries have examined the relationship between fiscal policy and various welfare indicators—such as income inequality, access to education, healthcare provision, and poverty alleviation—there remains a paucity of research focusing specifically on the Ghanaian context. In particular, few studies have comprehensively investigated the impact of taxation on income inequality, educational attainment, healthcare accessibility, and poverty reduction in Ghana. This lack of targeted research constitutes a notable gap in the literature and underscores the need for context-specific empirical analyses.

Moreover, the absence of up-to-date and reliable data on taxation and welfare outcomes in Ghana presents additional challenges for both researchers and policymakers. Existing studies often rely on outdated datasets, generalized cross-country comparisons, or aggregated measures that fail to capture the nuanced effects of different tax types on specific welfare indicators. For example, the differential impact of direct taxes (such as personal income tax, corporate tax, and capital gains tax) versus indirect taxes on income redistribution, educational financing,

or healthcare provision remains underexplored. Similarly, the indices and proxies commonly employed to measure welfare outcomes may not encompass all relevant dimensions, potentially overlooking critical aspects of well-being and socio-economic equity.

Consequently, this study seeks to address these gaps by undertaking a comprehensive empirical analysis of taxation in Ghana and its influence on key welfare indicators. By examining both direct and indirect taxes alongside measures of income inequality, educational outcomes, healthcare access, and poverty reduction, this research aims to provide robust, context-specific evidence to inform policy decisions. The study also emphasizes the need for improved data collection and methodological rigor to capture the full spectrum of taxation's socio-economic impacts, thereby contributing to more effective and equitable fiscal policy formulation.

Chapter 3 Research Methodology

3.1 Introduction

The central focus of this study is to analyze the impact of taxation on welfare in Ghana, as outlined in the preceding sections. This chapter details the research methodology employed to investigate the relationship between taxation policies and welfare in Ghana. It encompasses the research design and strategy, data sources, model specification, definition of variables, apriori expectations, estimation techniques, and data analysis procedures, and concludes with a summary of the chapter.

3.2 Research Design and Strategy

This study adopts a positivist research philosophy. Positivism emphasizes the importance of empirical evidence and seeks to explain phenomena through systematic observation and measurement (Noriey, 2022). In the context of this research, a positivist approach is essential for objectively assessing the impact of taxation on welfare, specifically its effects on welfare indicators such as income inequality, education, healthcare, and poverty reduction in Ghana, relying on quantifiable data and rigorous statistical analysis.

This research employs a quantitative research design, which is well-suited for analyzing numerical data to establish relationships and detect trends (Hadista, Suyadi, & Aisyah, 2021). Through quantitative analysis, this study aims to provide a precise and objective understanding of how

taxation policies influence welfare outcomes over a defined period.

The research strategy employed is Time Series Analysis. This approach involves collecting observations over successive time intervals, enabling the examination of correlations and trends across different income brackets and tax policies (Auer, Cavanaugh, Petor, Kinney, & Landers, 2022). This methodology is particularly effective for evaluating the temporal impact of taxation on income inequality. The time horizon for this study spans 40 years, from 1983 to 2022, allowing for a comprehensive assessment of taxation policies on income inequality within a relatively stable macroeconomic environment, while capturing variations resulting from policy changes and economic fluctuations.

3.3 Data Source

The study utilizes annual time-series data spanning from 1983 to 2022 (40 observations) to examine the impact of taxation on welfare in Ghana. Time-series data refers to a sequence of observations collected over successive periods, which permits the analysis of temporal dynamics and trends in economic and social indicators over time. To ensure comprehensive coverage and accuracy, this study draws on multiple reliable data sources, including national and international statistical repositories. A primary source is the **Ghana Statistical Service (GSS) StatsBank**, which provides official aggregated time-series data on key indicators such as national accounts, demographic characteristics, education statistics, health outcomes, poverty measures, and labor and income statistics for Ghana. StatsBank allows users to build customized tables and download historical datasets necessary for longitudinal analysis (Ghana Statistical Service, 2023). In addition, the **World Bank World Development Indicators (WDI)** database offers standardized time-series data covering measures such as GDP per capita, poverty headcount ratios, Gini coefficients, foreign direct investment inflows, and education and health outcomes, facilitating international comparability and econometric modeling over extended periods (World Bank, 2023). Tax and non-tax revenue data were obtained from the **UNU-WIDER Government Revenue Dataset (GRD)** (UNU-WIDER, 2023), which compiles internationally comparable government revenue statistics from national sources and the International Monetary Fund. Direct taxes, indirect taxes, total taxes, and non-tax revenue are all expressed as a

percentage of GDP. Measuring taxes relative to GDP captures the tax burden on the economy, which is the measure most commonly used in studies of taxation and welfare, and removes the effect of inflation on nominal revenue figures: in nominal cedis, direct tax revenue rose more than 100,000-fold between 1983 and 2022, mainly because of price increases rather than changes in the tax burden. Furthermore, specific welfare-related statistics, such as demographic and health indicators, are available through the **Ghana Demographic and Health Survey (GDHS)**, which provides national time-series data on health access, mortality, fertility, and service utilization, useful for analyzing changes in healthcare access over time (Ghana Demographic and Health Survey (GDHS), 2023). For income inequality measures, the Gini index for Ghana is available from datasets compiled by the **World Bank and its partner databases**, which aggregate annual series on inequality indicators derived from household surveys and national sources (Federal Reserve Bank of St. Louis (FRED), 2023).

3.3.1 Treatment of Missing Data

Three of the variables used in the models are not available for every year of the study period. The Gini coefficient (GINI) and the poverty headcount ratio (POVH) are derived from the Ghana Living Standards Survey (GLSS) and are therefore available only for the survey years 1987, 1988, 1991, 1998, 2005, 2012, and 2016, that is, for 7 of the 40 years. Inequality and poverty statistics for Ghana are generally available only at such irregular intervals, because they depend on the timing of national household surveys (Atta-Ankomah, Lambon-Quayefio, & Osei, 2023). Secondary school enrollment (SSER) is missing for eight years (1992–1995, 1997–1999, and 2011). All other variables are available for every year from 1983 to 2022.

The missing values were filled by linear interpolation. For a year t lying between two observed years $t_1 < t < t_2$, the interpolated value is

$$y_t = y_{t_1} + \frac{t - t_1}{t_2 - t_1}\,(y_{t_2} - y_{t_1}) \tag{3.1}$$

which assumes that the variable changes by an equal amount in each year between two observations. For example, the poverty headcount was 55.0% in 1998 and 42.5% in 2005, so it is

assumed to fall by $(55.0 - 42.5)/7 \approx 1.79$ percentage points in each year between 1998 and 2005. For years before the first observation (1983–1986 for GINI and POVH) and after the last observation (2017–2022), where interpolation is not possible, the nearest observed value was carried backward or forward. Table 3.1 summarises the observed and interpolated years for each variable.

Table 3.1: Observed and Interpolated Years of Variables with Missing Data

Variable	**Observed years**	**Observed**	**Filled years (method)**
GINI	1987, 1988, 1991, 1998, 2005, 2012, 2016 (GLSS rounds)	7	23 interpolated; 1983–1986 and 2017–2022 carried from nearest survey
POVH	1987, 1988, 1991, 1998, 2005, 2012, 2016 (GLSS rounds)	7	23 interpolated; 1983–1986 and 2017–2022 carried from nearest survey
SSER	1983–1991, 1996, 2000–2010, 2012–2022	32	1992–1995, 1997–1999, 2011 (interpolated)
All other variables	1983–2022	40	None

Linear interpolation was chosen because it is simple and transparent, uses only the observed values of each variable, and does not introduce fluctuations that are not present in the data. Its main limitation is that it removes the year-to-year variation between survey years, so that the interpolated GINI and POVH series are smoother than the true series and contain much less independent information than 40 annual observations would suggest. In particular, the interpolated poverty headcount behaves as an I(2) series in the unit root tests (Section 4.4). Results involving GINI and POVH, and especially the poverty model, should therefore be interpreted with caution.

3.4 Model Specification

This study employs a multiple regression framework to investigate the relationship between taxation (independent variables) and welfare indicators (dependent variables), namely income inequality, access to education, access to healthcare, and poverty reduction, while controlling for relevant covariates. Four models were specified to achieve the study objectives, and a supplementary model was specified to capture economic growth, an important channel through

which taxation may affect poverty. All tax variables are expressed as a percentage of GDP. The equations below give the long-run form of each relationship; Section 3.7.2 explains how each is estimated in dynamic ARDL form. Total taxes are not included in the same model as direct and indirect taxes because they are largely the sum of the two, which would cause severe multicollinearity.

3.4.1 The First Model: Taxation and Income Inequality

Income inequality is expressed as a function of taxation:

$$\text{Income Inequality} = f(\text{Taxation}, \text{Controls}) \tag{3.2}$$

The estimable model is:

$$\text{GINI}_t = \mu_0 + \mu_1 \text{DITA}_t + \mu_2 \text{IDTA}_t + \mu_3 \text{FDI}_t + \mu_4 \text{INF}_t + \varepsilon_t \tag{3.3}$$

Where:

- GINI = Gini coefficient
- DITA = Direct taxes (% of GDP)
- IDTA = Indirect taxes (% of GDP)
- FDI = Foreign direct investment, net inflows (% of GDP)
- INF = Inflation rate (%)
- μ_0 = Intercept; μ_1–μ_4 = Coefficients; ε_t = Error term

3.4.2 The Second Model: Taxation and Access to Education

Access to education is proxied by the secondary school enrollment rate (SSER):

$$\text{SSER}_t = b_0 + b_1\text{DITA}_t + b_2\text{GDPPC}_t + b_3\text{GEX}_t + u_t \tag{3.4}$$

Where:

- SSER = Secondary school enrollment rate (% gross)
- DITA = Direct taxes (% of GDP)
- GDPPC = GDP per capita
- GEX = Government expenditure (% of GDP)
- b_0 = Constant; b_1–b_3 = Coefficients; u_t = Error term

3.4.3 The Third Model: Taxation and Access to Healthcare

Access to healthcare is proxied by child immunization coverage (IMU). Immunization coverage reflects the reach of the primary health care system and the extent to which households are in contact with health services, and it is widely used as an indicator of access to health services. The model is:

$$\text{IMU}_t = \alpha_0 + \alpha_1\text{DITA}_t + \alpha_2\text{IDTA}_t + \alpha_3\text{NTRA}_t + \alpha_4\text{INF}_t + e_t \tag{3.5}$$

Where:

- IMU = Immunization coverage (% of children aged 12–23 months)
- DITA, IDTA = Direct and indirect taxes (% of GDP)
- NTRA = Non-tax revenue (% of GDP)
- INF = Inflation rate (%)
- α_0 = Intercept; α_1–α_4 = Coefficients; e_t = Error term

3.4.4 The Fourth Model: Taxation and Poverty Reduction

Following the growth–poverty literature, poverty is related to income per capita, inequality, and fiscal variables. Extending this relationship to include taxation and control variables gives:

$$\text{POVH}_t = \delta_0 + \delta_1 \text{DITA}_t + \delta_2 \text{GDPPC}_t + \delta_3 \text{GINI}_t + \delta_4 \text{GEX}_t + \delta_5 \text{AG}_t + \varepsilon_t \tag{3.6}$$

Where:

- POVH = Poverty headcount ratio (%)
- DITA = Direct taxes (% of GDP)
- GDPPC = GDP per capita
- GINI = Gini coefficient
- GEX = Government expenditure (% of GDP)
- AG = Agriculture value added (% of GDP)
- δ_0 = Intercept; δ_1–δ_5 = Coefficients; ε_t = Error term

3.4.5 Supplementary Model: Taxation and Economic Growth

Because economic growth is a major channel through which taxation can affect poverty, a supplementary model relates GDP per capita to total taxes:

$$\text{GDPPC}_t = \gamma_0 + \gamma_1 \text{TOTA}_t + \gamma_2 \text{FDI}_t + \gamma_3 \text{GEX}_t + \gamma_4 \text{INF}_t + \gamma_5 \text{AG}_t + v_t \tag{3.7}$$

where TOTA is total tax revenue (% of GDP), the other variables are as defined above, γ_0 is the intercept, γ_1–γ_5 are coefficients, and v_t is the error term.

3.5 Definition of Variables

To ensure clarity and consistency in the analysis, the variables employed in this study are defined in Table 3.2, together with their units of measurement and data sources.

Table 3.2: Definition of Variables Used in the Study

Variable	**Definition and measurement**	**Source**
GINI	Gini index of income inequality (0–100)	World Bank; FRED
SSER	Secondary school enrollment (% gross)	World Bank WDI
IMU	Immunization coverage (% of children aged 12–23 months)	World Bank WDI
POVH	Poverty headcount ratio (% of population)	World Bank WDI; GSS
DITA	Direct taxes (% of GDP)	UNU-WIDER GRD
IDTA	Indirect taxes (% of GDP)	UNU-WIDER GRD
TOTA	Total tax revenue (% of GDP)	UNU-WIDER GRD
NTRA	Non-tax revenue (% of GDP)	UNU-WIDER GRD
GDPPC	GDP per capita (constant US$)	World Bank WDI
GEX	General government final consumption expenditure (% of GDP)	World Bank WDI
FDI	Foreign direct investment, net inflows (% of GDP)	World Bank WDI
INF	Inflation, consumer prices (annual %)	World Bank WDI
AG	Agriculture, value added (% of GDP)	World Bank WDI

3.5.1 Control Variables Justification

In the regression models specified, control variables are incorporated to account for other factors that could influence welfare outcomes in Ghana, so that the estimated effects of taxation are not biased. In the income inequality model, foreign direct investment (FDI) is included because inflows of capital can create employment opportunities and influence wage structures, potentially affecting inequality, and inflation (INF) controls for price changes that erode real incomes, particularly those of poorer households. In the education model, GDP per capita (GDPPC) captures the overall economic capacity of households and government to finance education, and government expenditure (GEX) captures public spending that may support schooling. In the healthcare model, non-tax revenue (NTRA) controls for fiscal capacity beyond taxation, and inflation controls for the effect of rising prices on the affordability of health services.

In the poverty reduction model, GDP per capita captures the growth–poverty linkage, the Gini coefficient (GINI) accounts for inequality, recognizing that higher inequality may hinder poverty reduction even when average incomes rise, government expenditure captures fiscal spending on social programs, and agriculture value added (AG) reflects the contribution of agriculture to livelihoods, particularly in rural areas where poverty is concentrated. In the supplementary growth model, FDI, government expenditure, inflation, and agriculture capture the main drivers of output identified in the growth literature. Poverty is measured by the poverty headcount ratio, the proportion of the population living below the poverty line.

3.6 Apriori Expectations

Based on theoretical and empirical literature, the following expectations are formulated for this study. Indirect taxes, such as value-added tax (VAT), customs duties, and excise duties, are expected to have a **positive relationship with income inequality** ($\mu_2 > 0$) due to their regressive nature, which disproportionately affects lower-income households. Conversely, direct taxes, including personal income tax, corporate tax, and capital gains tax, are anticipated to exhibit a **negative relationship with income inequality** ($\mu_1 < 0$), reflecting their redistributive effect on income.

Regarding education, taxes are expected to positively affect enrollment ($b_1 > 0$), as fiscal revenues provide resources for public education programs. For healthcare, the expectation is similarly **positive** ($\alpha_1, \alpha_2 > 0$), as taxation enables government spending on hospitals, clinics, immunization programmes, and health insurance, thereby improving access to health services.

In terms of poverty reduction, direct taxes are anticipated to reduce poverty ($\delta_1 < 0$) by providing revenue for social protection and welfare programs. Higher GDP per capita ($\delta_2 < 0$) and government expenditure ($\delta_4 < 0$) are expected to reduce poverty, while higher inequality is expected to increase it ($\delta_3 > 0$). For the supplementary growth model, the effect of total taxes on GDP per capita (γ_1) is ambiguous: taxes can finance productive public investment, but high taxes may also discourage private investment.

These expectations serve as a benchmark for evaluating empirical results, guiding interpretation

of the relationship between taxation and key welfare indicators.

3.7 Estimation Technique, Strategy, and Procedure

The estimation followed four steps: (1) testing each variable for a unit root to establish its order of integration; (2) testing for a long-run (cointegrating) relationship using the Autoregressive Distributed Lag (ARDL) bounds testing approach; (3) estimating the short-run and long-run effects of taxation on each welfare indicator; and (4) checking the estimated models with diagnostic and stability tests. Each step is described below.

3.7.1 Unit Root Test

Macroeconomic time series such as tax revenue and GDP per capita usually trend over time. Regressing one trending (non-stationary) series on another can produce a high R^2 and significant coefficients even when the variables are unrelated, a problem known as spurious regression. To avoid this, the stationarity of each variable was examined using the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979), based on the regression:

$$\Delta y_t = \alpha + \delta t + \gamma y_{t-1} + \sum_{i=1}^{k} \beta_i \Delta y_{t-i} + \varepsilon_t \tag{3.8}$$

where $\Delta y_t = y_t - y_{t-1}$ is the first difference of the series, t is a time trend, k is the number of lagged differences included to remove serial correlation in the error term, and ε_t is the error term. The null hypothesis is $H_0 : \gamma = 0$ (the series has a unit root and is non-stationary) against the alternative $H_1 : \gamma < 0$ (the series is stationary). The test was conducted first at levels and then at first difference. In both cases the test regression includes an intercept, a linear trend, and one lagged difference ($k = 1$), and p-values are computed from the response-surface critical values of MacKinnon (1996). A variable that is stationary at levels is integrated of order zero, I(0); a variable that becomes stationary only after first differencing is integrated of order one, I(1). Establishing the order of integration matters because the ARDL bounds test is valid for

a mixture of I(0) and I(1) variables but not for variables that are I(2).

3.7.2 The ARDL Model

The study adopted the ARDL approach developed by Pesaran, Shin, and Smith (2001). The approach was preferred over the Johansen cointegration and Vector Error Correction Model (VECM) framework for three reasons. First, it can be applied whether the regressors are I(0), I(1), or a mixture of both, which suits the mixed results of the unit root tests. Second, it performs well in small samples, which is important given the 38 annual observations available for this study. Third, a single equation provides both the short-run dynamics and the long-run relationship for each welfare indicator. The ARDL approach has also been widely applied to Ghanaian annual time series of similar length. Adu et al. (2023) used an asymmetric (nonlinear) ARDL model on Ghanaian data for 1983–2016, the same starting year as this study, to examine the effects of tax systems on poverty and inequality. Ofori, Obeng, and Armah (2018) applied the ARDL approach to examine the effect of exchange rate volatility on tax revenue in Ghana, and Boachie (2015) used the ARDL bounds test on annual data for 1982–2012 to examine the effect of health on economic growth in Ghana. These studies show that the ARDL approach is well suited to the short annual series available for Ghana and to questions linking fiscal variables with welfare outcomes. The general ARDL$(p, q_1, \ldots, q_k)$ model is specified as:

$$y_t = \alpha_0 + \sum_{i=1}^{p} \phi_i \, y_{t-i} + \sum_{j=1}^{k} \sum_{l=0}^{q_j} \theta_{jl} \, x_{j,t-l} + \varepsilon_t \tag{3.9}$$

where y_t is the welfare indicator (GINI, SSER, IMU, or POVH), $x_{j,t}$ are the tax and control variables specified in Section 3.4, p is the number of lags of the dependent variable, q_j is the number of lags of regressor j, and ε_t is the error term. The coefficients on the current values of the regressors (θ_{j0}) capture the short-run effects, while the coefficients on their lags ($\theta_{j1}, \theta_{j2}, \ldots$) capture the lagged effects. Given the annual frequency of the data and the small sample, a maximum of two lags was allowed for each variable. Every combination of lag lengths (one or two lags of the dependent variable and zero to two lags of each regressor) was estimated, and the combination with the lowest Akaike Information Criterion (AIC) was selected. All regressors

were retained in every model so that the effect of each tax variable could be assessed. The first two years of data (1983 and 1984) serve as initial values for the lags, so every model is estimated on the same 38 annual observations (1985–2022). The models were estimated by Ordinary Least Squares (OLS).

3.7.3 Bounds Test for Cointegration

Cointegration means that variables which individually trend over time move together in a stable long-run relationship. To test for cointegration, the ARDL model is rewritten in the following error correction form:

$$\Delta y_t = \alpha_0 + \lambda_0\, y_{t-1} + \sum_{j=1}^{k} \lambda_j\, x_{j,t-1} + \sum_{i=1}^{p-1} \phi_i^*\, \Delta y_{t-i} + \sum_{j=1}^{k} \sum_{l=0}^{q_j-1} \theta_{jl}^*\, \Delta x_{j,t-l} + \varepsilon_t \tag{3.10}$$

The null hypothesis of no long-run relationship, $H_0 : \lambda_0 = \lambda_1 = \cdots = \lambda_k = 0$, is tested against the alternative that at least one λ is non-zero using an F-test. The computed F-statistic is compared with two sets of critical values: a lower bound, which assumes all regressors are I(0), and an upper bound, which assumes all regressors are I(1). If the F-statistic exceeds the upper bound, the null hypothesis is rejected and cointegration is confirmed; if it falls below the lower bound, there is no cointegration; and if it lies between the two bounds, the test is inconclusive. The test is specified with an unrestricted intercept and no trend (Case III of Pesaran et al. (2001)). Because the sample is small, the asymptotic critical values of Pesaran et al. (2001) can be misleading. Following the reasoning of P. K. Narayan (2005), who showed that small-sample critical values are considerably higher than asymptotic ones, the critical values in this study were simulated for the exact sample size ($n = 38$) and lag structure of each model, using 20,000 replications.

3.7.4 Long-Run and Short-Run (Error Correction) Estimates

Where cointegration is confirmed, the long-run coefficient of each regressor is obtained from the error correction form as:

$$\beta_j = -\frac{\lambda_j}{\lambda_0} \tag{3.11}$$

which measures the total effect of a permanent one-unit change in x_j on y after all adjustments have taken place. The short-run dynamics are then estimated using the Error Correction Model (ECM):

$$\Delta y_t = \alpha_0 + \sum_{i=1}^{p-1} \phi_i^* \, \Delta y_{t-i} + \sum_{j=1}^{k} \sum_{l=0}^{q_j-1} \theta_{jl}^* \, \Delta x_{j,t-l} + \psi \, ECT_{t-1} + \varepsilon_t \tag{3.12}$$

where ECT_{t-1} is the error correction term, the lagged deviation of y from its long-run equilibrium. Its coefficient, ψ, is the speed of adjustment: it is expected to be negative and statistically significant, and it shows the proportion of any deviation from the long-run equilibrium that is corrected within one year.

3.7.5 Diagnostic and Stability Tests

To confirm that the estimated models are reliable, the residuals of each model were subjected to diagnostic tests. The Breusch-Godfrey Lagrange Multiplier (LM) test (Breusch, 1978; Godfrey, 1978) was used to test for serial correlation, the Breusch-Pagan test (Breusch & Pagan, 1979) for heteroskedasticity, and the Jarque-Bera test (Jarque & Bera, 1980) for normality of the residuals. In each case, the null hypothesis is that the assumption holds (no serial correlation, constant variance, and normally distributed residuals), so a p-value above 0.05 indicates that the model passes the test. Finally, the stability of the estimated coefficients over the sample period was examined using the cumulative sum (CUSUM) test (Brown, Durbin, & Evans, 1975); the coefficients are judged stable if the CUSUM plot remains within the 5% significance bounds.

3.8 Data Analysis and Techniques

The data analysis proceeded in stages. First, descriptive statistics (mean, standard deviation, median, minimum, maximum, skewness, and kurtosis) were computed to summarise the characteristics of the variables over the 1983–2022 period. Second, a correlation matrix, trend plots, and scatter plots were used to explore the behaviour of the key variables over time and the preliminary relationships between them. Third, the ADF unit root test was applied to each variable at levels and at first difference. Fourth, separate ARDL models were estimated for each of the four welfare indicators, following the specific objectives of the study: income inequality (Objective 1), access to education (Objective 2), access to healthcare (Objective 3), and poverty reduction (Objective 4), together with a supplementary model of GDP per capita. For each model, the short-run, lagged, and long-run effects of taxation were obtained. Finally, the diagnostic and stability tests described above were conducted to validate the models before the results were interpreted.

3.8.1 Chapter Summary

This chapter presented the research methodology employed to examine the impact of taxation on welfare in Ghana. A positivist research philosophy guided the study, underpinned by a quantitative research design using annual time series data spanning 1983 to 2022 (40 observations, of which 38 are used in estimation after allowing for two lags). Data were drawn from the Ghana Statistical Service, the World Bank World Development Indicators, and the UNU-WIDER Government Revenue Dataset, with missing years in the Gini coefficient, poverty headcount, and secondary school enrollment filled by linear interpolation, covering taxation variables, welfare indicators, and economic control variables. Four models were specified to investigate the relationships between taxation and income inequality, education, healthcare, and poverty reduction, with clear definitions of all key variables and apriori expectations grounded in theory and empirical literature. The estimation procedure involved the Augmented Dickey-Fuller unit root test, the ARDL bounds testing approach to cointegration, the estimation of short-run and long-run effects using the ARDL and error correction framework, and diagnostic and stability tests.

This methodological framework provides the foundation for the empirical results presented in the next chapter.

Chapter 4 Results

4.1 Introduction

This chapter presents the empirical results of the investigation into the relationship between taxation and welfare outcomes in Ghana. The analysis systematically addresses each research objective by examining how tax structures influence income inequality, educational access, healthcare provision, and poverty reduction. Descriptive statistics provide an overview of the data characteristics, followed by trend analysis of key variables and unit root tests to establish stationarity properties. The chapter subsequently presents ARDL model estimations, exploring both short-run dynamics and long-run equilibrium relationships, concluding with diagnostic tests to validate the robustness of the findings.

4.1.1 Descriptive Statistics

Table 4.1 presents the summary statistics for the variables employed in the study, covering the period 1983 to 2022 (40 annual observations, including the interpolated values described in Section 3.3.1). The mean Gini coefficient of 40.59 indicates moderate income inequality in Ghana, with relatively low variability (SD = 2.85). Poverty headcount (POVH) averaged 44.15% of the population, ranging from 25.20% to 64.20%. Secondary school enrollment (SSER) averaged 44.50%, rising from a low of 29.83% to 75.31%, while immunization coverage (IMU) averaged 74.65%, ranging from 10% to 95%. Total tax revenue (TOTA) averaged 8.24% of GDP, of which indirect taxes (IDTA, 5.34% of GDP) contributed considerably more than direct taxes (DITA, 2.78% of GDP), confirming the heavy reliance of Ghana's tax system on indirect taxes. Non-

tax revenue (NTRA) averaged only 1.43% of GDP. GDP per capita averaged $1,231.57 (SD = 420.98), and government expenditure (GEX) averaged 9.79% of GDP. Inflation (INF) displays extreme volatility (SD = 20.57) and high kurtosis (10.94), reflecting periods of macroeconomic instability, including the maximum of 122.87% recorded in 1983. These statistics provide essential context for understanding the welfare landscape within which Ghana's tax policies operate.

Table 4.1: Descriptive Statistics for Selected Variables

Variable	Mean	SD	Median	Min	Max	Range	Skew	Kurtosis
GINI	40.59	2.85	41.84	35.30	43.50	8.20	-0.71	-0.93
POVH	44.15	14.79	46.96	25.20	64.20	39.00	-0.17	-1.70
SSER	44.50	14.92	37.43	29.83	75.31	45.48	0.82	-0.97
IMU	74.65	22.10	82.00	10.00	95.00	85.00	-1.50	1.53
DITA	2.78	1.66	2.57	0.42	6.23	5.81	0.47	-1.04
IDTA	5.34	1.56	5.39	1.59	8.44	6.85	-0.24	-0.46
TOTA	8.24	3.08	8.19	2.01	13.29	11.27	-0.05	-1.14
NTRA	1.43	1.06	1.11	0.26	4.03	3.76	0.77	-0.73
GDPPC	1231.57	420.98	1062.86	702.42	2046.01	1343.58	0.63	-1.11
GEX	9.79	2.33	9.60	5.86	15.31	9.45	0.47	-0.65
FDI	3.19	2.84	2.17	0.05	9.47	9.42	0.67	-0.80
INF	23.64	20.57	17.30	4.87	122.87	118.01	2.91	10.94
AG	33.68	11.28	35.52	17.32	59.73	42.41	0.18	-1.01

Figure 4.1 presents the pairwise correlation coefficients among the variables. GDP per capita is strongly negatively correlated with poverty headcount ($r = -0.94$), consistent with the growth–poverty linkage. All three tax ratios are strongly and positively correlated with GDP per capita (for example, $r = 0.97$ for direct taxes), reflecting the expansion of the tax base as the economy has grown. The tax ratios are also positively correlated with the Gini coefficient ($r = 0.84$ for direct taxes and $r = 0.88$ for indirect taxes) and with immunization coverage ($r = 0.75$ and $r = 0.82$), and negatively correlated with poverty (r between -0.77 and -0.95). Government expenditure ($r = 0.45$) and inflation ($r = 0.44$) are positively correlated with poverty, while inflation is negatively correlated with inequality ($r = -0.51$). Agriculture's share of GDP is strongly negatively correlated with most variables (for example, $r = -0.94$ with total taxes), reflecting the structural transformation of the economy away from agriculture.

Because many of these variables trend over time, and because GINI and POVH are interpolated between survey years, high correlations in levels largely reflect shared long-run trends rather

than causal relationships. For example, the positive correlation between taxes and inequality does not imply that taxes increase inequality; both rose over the study period. The unit root tests and ARDL models in the following sections are designed to separate genuine relationships from such common trends.

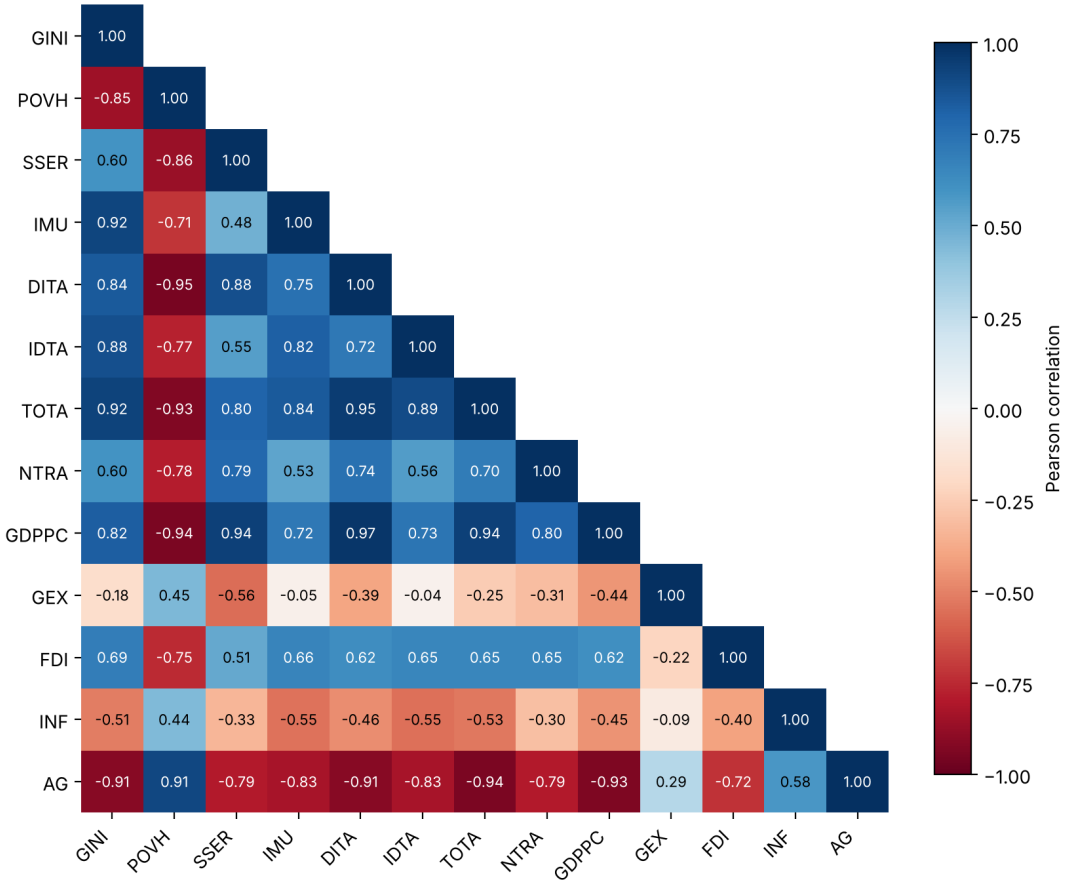


Figure 4.1: Correlation Matrix of Selected Economic and Social Variables

4.2 Trend Analysis of Key Variables

Figure 4.2 illustrates the temporal dynamics of secondary school enrollment (SSER), government expenditure (GEX), and inflation (INF) in Ghana from 1983 to 2022. Secondary school enrollment exhibits a pronounced upward trajectory, increasing from approximately 30% in the early 2000s to over 75% by 2022. This monotonic increase reflects the cumulative impact of educational policy interventions, including the Free Compulsory Universal Basic Education (FCUBE) program and the more recent Free Senior High School policy, which progressively

expanded access to secondary education. Government expenditure as a percentage of GDP displays moderate fluctuation around the mean of 9.79%, with notable peaks corresponding to election years and periods of fiscal expansion. The relative stability of GEX suggests consistent public sector commitment to welfare-related spending, though the magnitude of fluctuation increased post-2010. Inflation demonstrates extreme volatility: it peaked at 122.9% in 1983, spiked again to 59.5% in 1995 and to about 40% in 2000–2001, and rose to 31.3% in 2022. These inflationary episodes represent significant macroeconomic shocks that erode household purchasing power and potentially offset welfare gains from educational expansion and government spending. The juxtaposition of rising educational attainment against inflationary instability provides critical context for understanding the complex welfare landscape in Ghana.

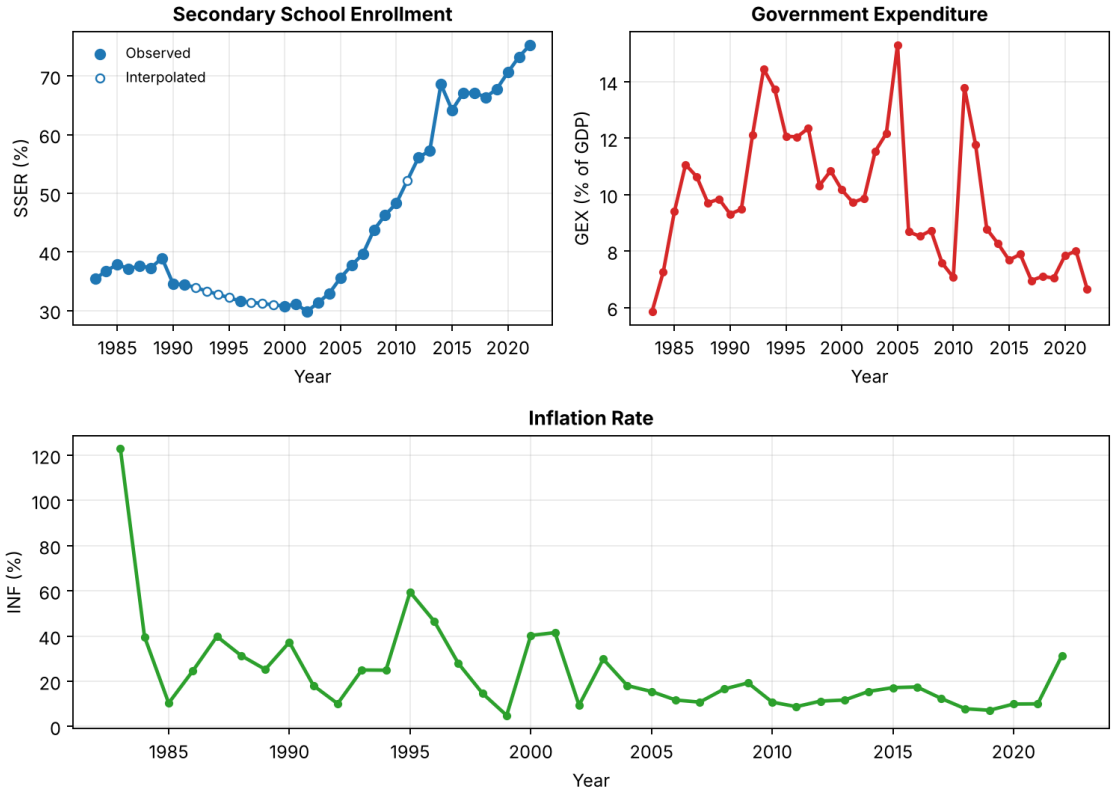


Figure 4.2: Time Series of Education, Government Expenditure, and Inflation Rate (for SSER, filled markers are observed and hollow markers interpolated values)

Figure 4.3 depicts the evolution of income inequality (GINI), poverty headcount (POVH), and GDP per capita (GDPPC) in Ghana over the 1983–2022 period. GDP per capita exhibits a consistent upward trajectory, increasing from approximately $700 at the start of the sample period to over $2,000 by 2022, reflecting sustained economic growth driven by macroeconomic

stability, commodity exports, and service sector expansion. This growth pattern aligns with Ghana's transition to lower-middle-income status in 2010. Poverty headcount, observed in the seven GLSS survey years, rose from 58.4% in 1987 to a peak of 64.2% in 1991 and then fell steadily to 42.5% in 2005, 25.7% in 2012, and 25.2% in 2016; the values between and after the survey years are interpolated, as shown by the hollow markers. This dramatic reduction corresponds with the growth-poverty elasticity literature, indicating that Ghana's growth episode has been broadly pro-poor, supported by social protection programs including the Livelihood Empowerment Against Poverty (LEAP) initiative and the National Health Insurance Scheme (NHIS). Income inequality, measured by the Gini coefficient, displays a more complex pattern. Inequality rose steadily from 35.3 in the mid-1980s to 42.8 in 2005, remained broadly stable until about 2012, and then rose to 43.5 by 2016, where it has since remained. The divergence between declining poverty and rising inequality post-2010 suggests that while growth has lifted households above the poverty line, the benefits have been unevenly distributed, with higher-income groups capturing a disproportionate share of economic gains—a pattern consistent with the Kuznets hypothesis in middle-income transitioning economies. This simultaneous reduction in poverty and increase in inequality presents a critical welfare paradox with significant implications for tax policy design.

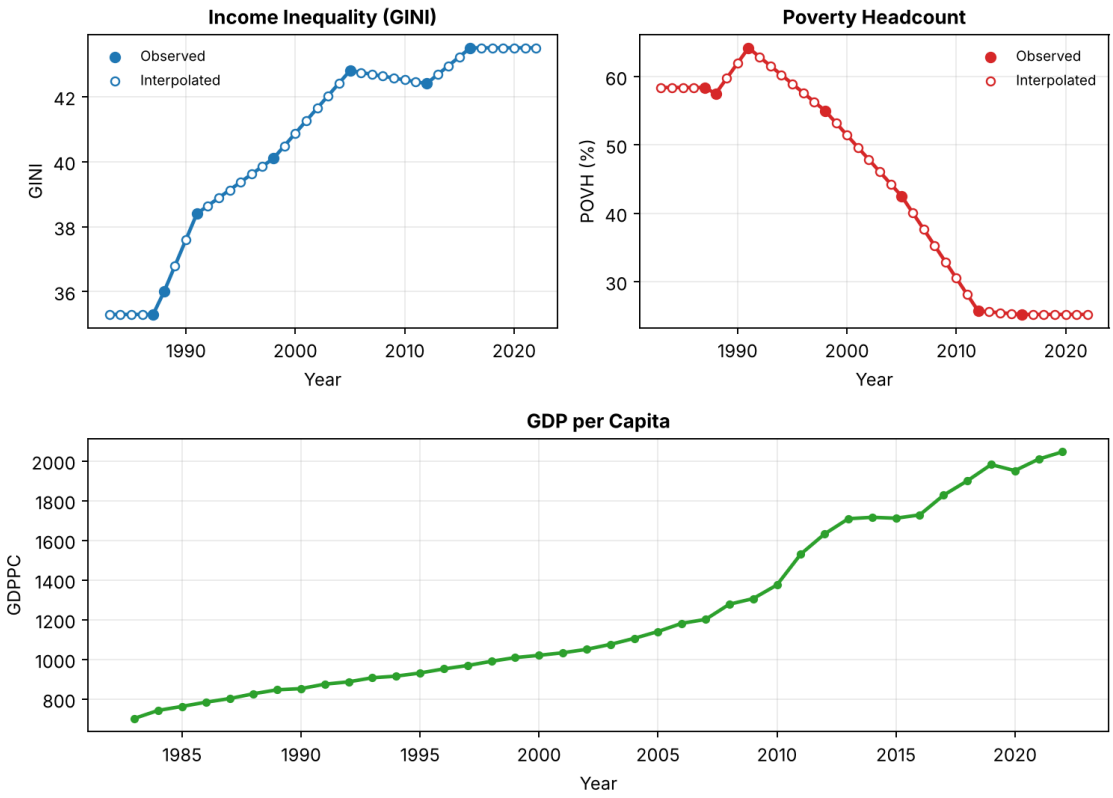


Figure 4.3: Time Series of Income Inequality, Poverty Headcount, and GDP per Capita (for GINI and POVH, filled markers are observed survey values and hollow markers interpolated values)

4.3 Exploratory Relationship Analysis

Figure 4.4 shows four bivariate relationships. First, GDP per capita is strongly negatively related to poverty headcount ($r = -0.94$): poverty is lower in years with higher real incomes, consistent with the growth–poverty literature, although because poverty is interpolated between survey years this association mainly reflects the long-run downward trend in poverty and upward trend in income. Second, secondary school enrollment is positively related to the Gini coefficient ($r = 0.60$). Because both enrollment and inequality rose over the study period, this association reflects shared trends rather than evidence that education increases inequality. Third, government expenditure is positively related to poverty ($r = 0.45$): expenditure as a share of GDP was higher in the 1990s and early 2000s, when poverty was also higher, so this association reflects the timing of the two series rather than an effect of spending on poverty.

Fourth, inflation is positively associated with poverty ($r = 0.44$), as both were higher in the 1980s and 1990s, although the relationship is influenced by extreme observations, particularly the 1983 hyperinflationary episode (122.9%). These bivariate patterns are only descriptive; the ARDL models below control for other variables and for the dynamics of each series.

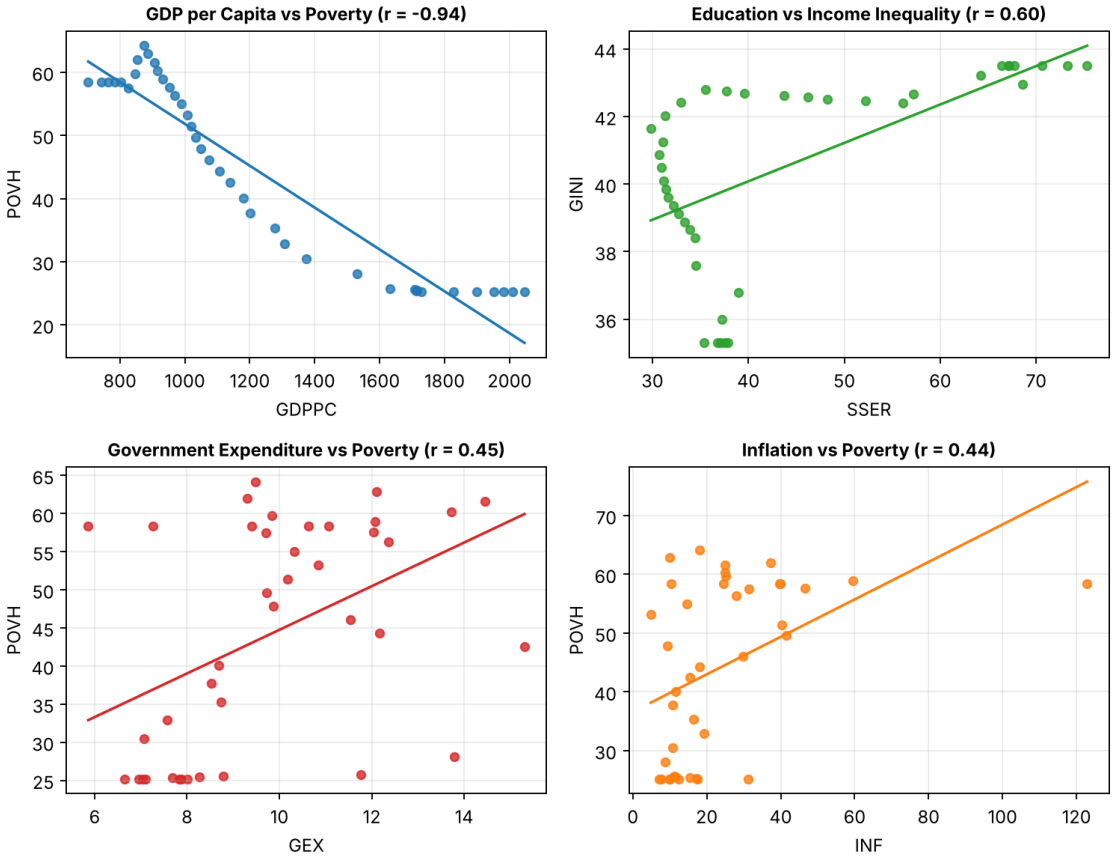


Figure 4.4: Scatter Plots Depicting Key Socioeconomic Relationships: GDP per Capita vs Poverty, Secondary School Enrollment vs Income Inequality, Government Expenditure vs Poverty, and Inflation Rate vs Poverty.

4.4 Unit Root Test Results

Before estimating the ARDL models, the stationarity properties of all variables were examined using the Augmented Dickey-Fuller (ADF) test, with the null hypothesis that the series contains a unit root (non-stationary). The test was first conducted on the variables at levels (Table 4.2) and then on their first differences (Table 4.3). Each test includes an intercept, a linear trend, and one lagged difference, and p-values are based on MacKinnon (1996).

Table 4.2 presents the ADF test results at levels. Immunization coverage (IMU), total taxes (TOTA), government expenditure (GEX), inflation (INF), and agriculture value added (AG) are stationary at the 5% significance level and are therefore integrated of order zero, I(0). These variables exhibit mean-reverting behavior, implying that shocks to them are temporary. The remaining variables—the Gini coefficient (GINI), secondary school enrollment (SSER), poverty headcount (POVH), direct taxes (DITA), indirect taxes (IDTA), non-tax revenue (NTRA), GDP per capita (GDPPC), and foreign direct investment (FDI)—have p-values above 0.05 at levels, so the null hypothesis of a unit root cannot be rejected. These variables were therefore differenced once and tested again.

Table 4.2: ADF Unit Root Test Results at Levels

Variable	ADF Statistic	p-value	Decision
GINI	-1.8543	0.6780	Non-stationary
SSER	-0.9122	0.9548	Non-stationary
IMU	-4.2505	0.0038	Stationary, I(0)
POVH	-2.3556	0.4035	Non-stationary
DITA	-2.4343	0.3616	Non-stationary
IDTA	-2.4887	0.3336	Non-stationary
TOTA	-4.2800	0.0034	Stationary, I(0)
NTRA	-2.0346	0.5825	Non-stationary
GDPPC	-1.3830	0.8658	Non-stationary
GEX	-3.8316	0.0150	Stationary, I(0)
FDI	-1.6214	0.7840	Non-stationary
INF	-5.8527	0.0000	Stationary, I(0)
AG	-4.1794	0.0048	Stationary, I(0)

Note: H_0: the series has a unit root. Decision based on the 5% significance level.

Table 4.3 presents the ADF test results at first difference. All variables except the poverty headcount are stationary at first difference at the 5% significance level, so the variables that were non-stationary at levels are integrated of order one, I(1). The tax variables are I(1) because they are measured as a share of GDP: when measured in nominal cedis, the tax revenue series grow more than 100,000-fold over the sample period and remain non-stationary even after first differencing.

The poverty headcount is the exception: it remains non-stationary at first difference and becomes stationary only after second differencing (ADF = -4.3035, p = 0.0031), so it behaves as an I(2) series. This is largely an artefact of the linear interpolation described in Section

3.3.1. Because POVH is observed in only seven survey years, the interpolated series consists of straight-line segments; its first difference is therefore constant within each segment and changes only at the survey years, which the ADF test cannot distinguish from a non-stationary series. As a result, the ARDL bounds test, which requires that no variable is I(2), is not valid for the poverty model, and its results are treated as indicative only. All other models satisfy the conditions for the ARDL bounds testing approach.

Table 4.3: ADF Unit Root Test Results at First Difference

Variable	ADF Statistic	p-value	Order of Integration
ΔGINI	-3.5040	0.0390	I(1)
ΔSSER	-3.7204	0.0210	I(1)
ΔIMU	-3.8313	0.0150	I(0)
ΔPOVH	-1.9393	0.6341	Non-stationary, I(2)[a]
ΔDITA	-4.6931	0.0007	I(1)
ΔIDTA	-4.0487	0.0075	I(1)
ΔTOTA	-4.9450	0.0003	I(0)
ΔNTRA	-4.0437	0.0076	I(1)
ΔGDPPC	-3.5397	0.0353	I(1)
ΔGEX	-5.6684	0.0000	I(0)
ΔFDI	-4.2587	0.0036	I(1)
ΔINF	-7.5730	0.0000	I(0)
ΔAG	-6.9440	0.0000	I(0)

Note: H_0: the series has a unit root. Δ denotes the first difference.

[a]Stationary at second difference; the I(2) behaviour reflects linear interpolation between seven survey years.

4.5 ARDL Model Estimation

This section presents the Autoregressive Distributed Lag (ARDL) estimates of the effects of taxation on welfare outcomes in Ghana. Because the variables are a mixture of I(0) and I(1) series (Tables 4.2 and 4.3), the ARDL bounds testing approach is appropriate for all models except the poverty model, whose dependent variable behaves as an I(2) series because of interpolation. Following the specific objectives of the study, the results are organised into four subsections: taxation and income inequality (Objective 1), taxation and access to education (Objective 2), taxation and access to healthcare (Objective 3), and taxation and poverty reduction (Objective 4). Table 4.4 first reports the bounds test for cointegration for every model. Under each objective, the short-run and lagged effects are then presented, followed by the long-

run coefficients and the error correction term (ECT). All models are estimated on 38 annual observations (1985–2022), and the lag structure of each model was selected by the AIC.

Table 4.4: ARDL Bounds Test for Cointegration

Model	**ARDL order**	**F-statistic**	k	**5% critical values**		**Decision**
				I(0)	I(1)	
Income inequality (GINI)	ARDL(2, 0, 1, 0, 2)	2.032	4	3.263	4.661	No cointegration
Education (SSER)	ARDL(2, 2, 2, 0)	4.289	3	3.531	4.997	Inconclusive[a]
Healthcare (IMU)	ARDL(2, 1, 2, 2, 2)	7.013	4	3.122	4.775	Cointegrated
Poverty (POVH)	ARDL(1, 0, 0, 2, 0, 2)	24.460	5	2.981	4.412	Not valid[b]
Growth (GDPPC)	ARDL(1, 1, 2, 2, 0, 2)	6.876	5	3.007	4.490	Cointegrated

Note: Case III (unrestricted intercept, no trend). Critical values simulated for the estimation sample ($n = 38$) and each model's lag structure (20,000 replications). k = number of regressors. [a]Above the 10% upper bound. [b]The F-statistic exceeds the upper bound, but the test is not valid because the interpolated poverty headcount behaves as an I(2) series (Section 4.4).

Table 4.4 shows that the F-statistic exceeds the 5% upper bound for the healthcare model (7.013 $>$ 4.775) and the growth model (6.876 $>$ 4.490). For both models, the F-statistic also exceeds the 1% upper bound, providing strong evidence of a long-run (cointegrating) relationship. For the income inequality model, the F-statistic (2.032) lies below the lower bound, so the null hypothesis of no long-run relationship cannot be rejected. For the education model, the F-statistic (4.289) lies between the 5% bounds and is therefore inconclusive, although it exceeds the 10% upper bound (4.156). For the poverty model, the F-statistic (24.460) is far above the upper bound, but the test is not valid because the interpolated poverty headcount behaves as an I(2) series; the very high F-statistic reflects the smoothness of the interpolated series rather than a genuine long-run relationship. Where there is no clear evidence of cointegration, the long-run coefficients are reported for completeness but should be interpreted with caution, and the analysis focuses on the short-run and lagged effects.

4.5.1 Taxation and Income Inequality

The first objective of the study was to assess the effect of taxes on income inequality in Ghana. Income inequality is measured by the Gini coefficient (GINI). The AIC selected an ARDL(2, 0, 1, 0, 2) model, with the regressors ordered as DITA, IDTA, FDI, and INF.

4.5.1.1 Short-Run Effects of Taxation on Income Inequality

Table 4.5 presents the short-run and lagged effects of taxation on income inequality. The model has high explanatory power (adjusted $R^2 = 0.9964$), largely because of the strong persistence of the Gini coefficient: its first lag is positive and highly significant (1.6616, $p < 0.001$) and its second lag is negative and significant (-0.7007, $p < 0.001$), indicating that inequality evolves slowly and responds mainly to its own past values.

Neither direct taxes nor indirect taxes have a statistically significant effect on income inequality. The coefficient of direct taxes (DITA) is positive but insignificant (0.0554, $p = 0.1460$), and the current and lagged coefficients of indirect taxes are also insignificant (0.0486, $p = 0.3041$; -0.0655, $p = 0.1894$). These results indicate that changes in the tax burden do not produce measurable changes in income distribution in the short run. Among the control variables, FDI is insignificant ($p = 0.9738$), and only the first lag of inflation is marginally significant (0.0046, $p = 0.0930$), suggesting that past inflation slightly raises inequality, possibly because price increases erode the real incomes of poorer households more than those of richer households.

Table 4.5: Short-Run and Lagged Effects of Taxation on Income Inequality

Variable	**Coefficient**	**Std. Error**	**p-value**
$GINI_{t-1}$	1.6616	0.1356	0.0000***
$GINI_{t-2}$	-0.7007	0.1430	0.0000***
DITA	0.0554	0.0371	0.1460
IDTA	0.0486	0.0464	0.3041
$IDTA_{t-1}$	-0.0655	0.0487	0.1894
FDI	0.000521	0.0157	0.9738
INF	0.000358	0.0027	0.8951
INF_{t-1}	0.0046	0.0026	0.0930*
INF_{t-2}	-0.0024	0.0018	0.1925
Constant	1.5335	0.9425	0.1149
R^2	0.9972		
Adjusted R^2	0.9964		
F-statistic	1124.70 ($p < 0.001$)		
Observations	38 (1985–2022)		

Dependent variable: GINI. ARDL(2, 0, 1, 0, 2) selected by AIC. *** p<0.01, ** p<0.05, * p<0.10.

4.5.1.2 Long-Run Effect of Taxation on Income Inequality

The bounds test (Table 4.4) found no long-run relationship between the tax variables and income inequality. Consistent with this, the error correction term is small and insignificant (-0.0391, $p = 0.1858$), and none of the long-run coefficients in Table 4.6 is significant: direct taxes (1.4181, $p = 0.1398$), indirect taxes (-0.4335, $p = 0.7502$), FDI ($p = 0.9735$), and inflation ($p = 0.5679$). Taken together, the short-run and long-run results show that taxation has no significant effect on income inequality in Ghana. This suggests that the tax system functions primarily as a revenue-raising instrument rather than a redistributive one, possibly because of the large informal sector, limited direct tax coverage, and the predominance of indirect taxes in the revenue mix.

Table 4.6: Long-Run Effects of Taxation on Income Inequality

Variable	**Coefficient**	**Std. Error**	**t-statistic**	**p-value**
DITA	1.4181	0.9332	1.520	0.1398
IDTA	-0.4335	1.3484	-0.322	0.7502
FDI	0.0133	0.3984	0.033	0.9735
INF	0.0650	0.1124	0.578	0.5679
Constant	39.2391	7.6569	5.125	0.0000***
ECT_{t-1} (speed of adjustment)	-0.0391	0.0288	-1.356	0.1858

Dependent variable: GINI. ARDL(2, 0, 1, 0, 2); observations: 38 (1985–2022). *** p<0.01, ** p<0.05, * p<0.10.

4.5.2 Taxation and Access to Education

The second objective of the study was to determine the effect of taxes on access to education in Ghana. Access to education is proxied by secondary school enrollment (SSER). The AIC selected an ARDL(2, 2, 2, 0) model, with the regressors ordered as DITA, GDPPC, and GEX.

4.5.2.1 Short-Run Effects of Taxation on Access to Education

Table 4.7 presents the short-run and lagged effects of taxation on secondary school enrollment. The model explains about 98% of the variation in enrollment (adjusted $R^2 = 0.9837$). Enrollment is highly persistent: its first lag (0.6332, $p < 0.001$) and second lag (0.3647, $p = 0.0205$)

are both positive and significant.

The contemporaneous effect of direct taxes is positive and marginally significant (2.4631, p = 0.0790), and the first lag is negative and marginally significant (−3.2677, p = 0.0645). The second lag of direct taxes is positive and highly significant (4.6006, p = 0.0015): a one percentage point increase in direct taxes as a share of GDP is associated with an increase of about 4.60 percentage points in secondary school enrollment two years later. Summing the three coefficients gives a cumulative effect of about 3.80 percentage points over two years. This delayed effect is consistent with fiscal implementation lags: tax revenues must be collected, appropriated through the budget, and allocated to the education sector before they translate into classrooms, teachers, and learning materials that raise enrollment.

Among the control variables, the second lag of GDP per capita is negative and significant (−0.0308, p = 0.0134), while its current value (p = 0.6211) and first lag (p = 0.1602) are insignificant. Government expenditure is insignificant (0.1080, p = 0.5869), suggesting that aggregate spending, which includes many non-education items, does not translate directly into enrollment gains.

Table 4.7: Short-Run and Lagged Effects of Taxation on Secondary School Enrollment

Variable	**Coefficient**	**Std. Error**	**p-value**
$SSER_{t-1}$	0.6332	0.1456	0.0002***
$SSER_{t-2}$	0.3647	0.1485	0.0205**
DITA	2.4631	1.3511	0.0790*
$DITA_{t-1}$	-3.2677	1.6977	0.0645*
$DITA_{t-2}$	4.6006	1.3092	0.0015***
GDPPC	-0.0064	0.0128	0.6211
$GDPPC_{t-1}$	0.0260	0.0180	0.1602
$GDPPC_{t-2}$	-0.0308	0.0117	0.0134**
GEX	0.1080	0.1965	0.5869
Constant	3.1156	3.3860	0.3654
R^2	0.9876		
Adjusted R^2	0.9837		
F-statistic	248.44 (p < 0.001)		
Observations	38 (1985–2022)		

Dependent variable: SSER. ARDL(2, 2, 2, 0) selected by AIC. *** p<0.01, ** p<0.05, * p<0.10.

4.5.2.2 Long-Run Effect of Taxation on Access to Education

The bounds test for the education model is inconclusive at the 5% level (F = 4.289). Moreover, the error correction term is almost exactly zero and insignificant (−0.0021, p = 0.9795), which indicates that enrollment does not adjust towards a long-run equilibrium with the tax variables. Because the long-run coefficients are obtained by dividing by this near-zero term, they are extremely large and imprecise (Table 4.8; all p-values are about 0.98) and should not be interpreted. The effect of direct taxes on education is therefore a delayed short-run effect, appearing two years after taxes are collected, rather than a stable long-run equilibrium relationship.

Table 4.8: Long-Run Effects of Taxation on Secondary School Enrollment

Variable	Coefficient	Std. Error	t-statistic	p-value
DITA	1838.0136	71261.8873	0.026	0.9796
GDPPC	-5.4164	211.3705	-0.026	0.9797
GEX	52.2988	2055.3954	0.025	0.9799
Constant	1508.5744	58202.8381	0.026	0.9795
ECT_{t-1} (speed of adjustment)	-0.0021	0.0797	-0.026	0.9795

Dependent variable: SSER. ARDL(2, 2, 2, 0); observations: 38 (1985–2022). *** p<0.01, ** p<0.05, * p<0.10.

4.5.3 Taxation and Access to Healthcare

The third objective of the study was to determine the effect of taxes on access to healthcare in Ghana. Access to healthcare is proxied by immunization coverage (IMU), the share of children aged 12–23 months who are immunized. The AIC selected an ARDL(2, 1, 2, 2, 2) model, with the regressors ordered as DITA, IDTA, NTRA, and INF.

4.5.3.1 Short-Run Effects of Taxation on Access to Healthcare

Table 4.9 presents the short-run and lagged effects of taxation on immunization coverage. The model explains about 94% of the variation in immunization coverage (adjusted $R^2 = 0.9422$), and the first lag of immunization coverage is positive and significant (0.4309, p = 0.0172).

Direct taxes have no significant short-run effect on immunization coverage, either contemporaneously (2.2749, p = 0.3513) or with a lag (−3.1569, p = 0.2172). Indirect taxes are insignificant contemporaneously (−0.0232, p = 0.9849) and at the first lag (−0.7907, p = 0.6235), but their second lag is positive and significant (2.9003, p = 0.0465): a one percentage point increase in indirect taxes as a share of GDP is associated with a 2.90 percentage point increase in immunization coverage two years later. Non-tax revenue shows a similar delayed pattern, with its second lag positive and marginally significant (2.7073, p = 0.0795). Inflation is insignificant at all lags.

Table 4.9: Short-Run and Lagged Effects of Taxation on Immunization Coverage

Variable	**Coefficient**	**Std. Error**	**p-value**
IMU_{t-1}	0.4309	0.1683	0.0172**
IMU_{t-2}	0.1938	0.1402	0.1797
DITA	2.2749	2.3934	0.3513
$DITA_{t-1}$	-3.1569	2.4908	0.2172
IDTA	-0.0232	1.2083	0.9849
$IDTA_{t-1}$	-0.7907	1.5901	0.6235
$IDTA_{t-2}$	2.9003	1.3814	0.0465**
NTRA	-0.0994	1.4996	0.9477
$NTRA_{t-1}$	-2.1049	1.8168	0.2580
$NTRA_{t-2}$	2.7073	1.4784	0.0795*
INF	0.0119	0.0701	0.8669
INF_{t-1}	-0.0850	0.0705	0.2397
INF_{t-2}	-0.0795	0.0520	0.1392
Constant	24.8871	6.1703	0.0005***
R^2		0.9625	
Adjusted R^2		0.9422	
F-statistic		47.43 (p < 0.001)	
Observations		38 (1985–2022)	

Dependent variable: IMU. ARDL(2, 1, 2, 2, 2) selected by AIC. *** p<0.01, ** p<0.05, * p<0.10.

4.5.3.2 Long-Run Effect of Taxation on Access to Healthcare

Unlike the other welfare indicators, healthcare access has a strong long-run relationship with taxation. The bounds F-statistic (7.013) exceeds the upper bound at both the 5% and 1% levels, and the error correction term in Table 4.10 is negative and highly significant (−0.3754, p < 0.001). This means that about 37.5% of any deviation of immunization coverage from its long-run equilibrium is corrected within one year, so that half of any deviation is eliminated in

about one and a half years.

In the long run, indirect taxes have a positive and significant effect on immunization coverage (5.5583, p = 0.0337): a sustained one percentage point increase in indirect taxes as a share of GDP raises immunization coverage by about 5.6 percentage points. Direct taxes (−2.3497, p = 0.4405), non-tax revenue (1.3399, p = 0.6715), and inflation (−0.4066, p = 0.1432) have no significant long-run effects. The positive role of indirect taxes is consistent with Ghana's health financing arrangements: the National Health Insurance Scheme (NHIS) is funded mainly through the National Health Insurance Levy, a 2.5% levy on goods and services, so revenue from indirect taxes is directly linked to the financing of health services.

Table 4.10: Long-Run Effects of Taxation on Immunization Coverage

Variable	**Coefficient**	**Std. Error**	**t-statistic**	**p-value**
DITA	-2.3497	2.9956	-0.784	0.4405
IDTA	5.5583	2.4676	2.252	0.0337**
NTRA	1.3399	3.1208	0.429	0.6715
INF	-0.4066	0.2686	-1.514	0.1432
Constant	66.3018	16.3427	4.057	0.0005***
ECT_{t-1} (speed of adjustment)	-0.3754	0.0752	-4.994	0.0000***

Dependent variable: IMU. ARDL(2, 1, 2, 2, 2); observations: 38 (1985–2022). *** p<0.01, ** p<0.05, * p<0.10.

4.5.4 Taxation and Poverty Reduction

The fourth objective of the study was to assess the effect of taxes on poverty reduction in Ghana. Poverty is measured by the poverty headcount ratio (POVH), which is observed in only seven survey years and interpolated for the remaining years (Section 3.3.1). The AIC selected an ARDL(1, 0, 0, 2, 0, 2) model, with the regressors ordered as DITA, GDPPC, GINI, GEX, and AG. Because the interpolated poverty headcount behaves as an I(2) series, the results in this subsection are indicative only. Because economic growth is an important channel through which taxation can affect poverty, this subsection also reports the supplementary growth model.

4.5.4.1 Short-Run Effects of Taxation on Poverty Reduction

Table 4.11 presents the short-run and lagged effects of taxation on poverty headcount. The model has very high explanatory power (adjusted $R^2 = 0.9988$), and the first lag of poverty is close to one (0.9937, $p < 0.001$). Both features largely reflect the smoothness of the interpolated poverty series, in which each year's value is close to the previous year's.

Direct taxes have no significant effect on poverty (-0.5049, $p = 0.1516$). Government expenditure has a negative and significant effect (-0.1172, $p = 0.0295$): a one percentage point increase in government expenditure as a share of GDP is associated with a reduction of about 0.12 percentage points in the poverty headcount, consistent with the role of public spending on social programs in reducing poverty. GDP per capita has a small positive and significant coefficient (0.0066, $p = 0.0013$), which is contrary to expectations; it likely reflects the fact that GDP per capita grew rapidly after 2012, while the interpolated poverty headcount was held constant at its 2016 survey value. The current Gini coefficient is positive and significant (1.3445, $p = 0.0336$), consistent with the view that higher inequality raises poverty, but its second lag is negative and significant (-1.5910, $p = 0.0433$). Agriculture's second lag is positive and significant (0.0843, $p = 0.0298$).

Table 4.11: Short-Run and Lagged Effects of Taxation on Poverty Headcount

Variable	**Coefficient**	**Std. Error**	**p-value**
$POVH_{t-1}$	0.9937	0.0248	0.0000***
DITA	-0.5049	0.3421	0.1516
GDPPC	0.0066	0.0019	0.0013***
GINI	1.3445	0.6007	0.0336**
$GINI_{t-1}$	-0.1291	1.1741	0.9133
$GINI_{t-2}$	-1.5910	0.7503	0.0433**
GEX	-0.1172	0.0510	0.0295**
AG	0.0831	0.0500	0.1079
AG_{t-1}	-0.0755	0.0497	0.1401
AG_{t-2}	0.0843	0.0367	0.0298**
Constant	5.2220	9.1806	0.5742
R^2	0.9991		
Adjusted R^2	0.9988		
F-statistic	3027.39 ($p < 0.001$)		
Observations	38 (1985–2022)		

Dependent variable: POVH. ARDL(1, 0, 0, 2, 0, 2) selected by AIC. *** p<0.01, ** p<0.05, * p<0.10.

4.5.4.2 Long-Run Effect of Taxation on Poverty Reduction

As explained above, the bounds test is not valid for the poverty model. Consistent with this, the error correction term is close to zero and insignificant (-0.0063, $p = 0.8003$), and the long-run coefficients in Table 4.12, which are obtained by dividing by this near-zero term, are very large and insignificant (all p-values are about 0.8). There is therefore no reliable evidence of a long-run relationship between taxation and poverty. Overall, taxation has no significant direct effect on poverty in Ghana, while government expenditure is associated with lower poverty in the short run; however, these conclusions are limited by the small number of poverty observations.

Table 4.12: Long-Run Effects of Taxation on Poverty Headcount

Variable	**Coefficient**	**Std. Error**	**t-statistic**	**p-value**
DITA	-79.5525	306.8924	-0.259	0.7974
GDPPC	1.0478	4.1782	0.251	0.8039
GINI	-59.1893	227.4081	-0.260	0.7966
GEX	-18.4597	75.3837	-0.245	0.8084
AG	14.4677	57.9242	0.250	0.8047
Constant	822.8254	3055.8188	0.269	0.7898
ECT_{t-1} (speed of adjustment)	-0.0063	0.0248	-0.255	0.8003

Dependent variable: POVH. ARDL(1, 0, 0, 2, 0, 2); observations: 38 (1985–2022). *** p<0.01, ** p<0.05, * p<0.10.

4.5.4.3 Taxation, Economic Growth and Poverty Reduction

Table 4.13 presents the supplementary model of GDP per capita, estimated as an ARDL(1, 1, 2, 2, 0, 2) with the regressors ordered as TOTA, FDI, GEX, INF, and AG. GDP per capita is highly persistent (first lag = 0.9352, $p < 0.001$). Total taxes have an insignificant contemporaneous effect (-5.5875, $p = 0.5079$), but their first lag is positive and significant (22.6738, $p = 0.0186$): a one percentage point increase in the tax-to-GDP ratio is associated with an increase of about US$22.67 in GDP per capita in the following year, consistent with tax revenues financing productive public investment. Foreign direct investment has a strong delayed effect, with its second lag positive and highly significant (12.8735, $p = 0.0031$), while the second lag of government expenditure is negative and marginally significant (-5.6087, $p = 0.0534$). Inflation and agriculture are insignificant.

Table 4.13: Short-Run and Lagged Effects of Taxation on GDP per Capita

Variable	Coefficient	Std. Error	p-value
$GDPPC_{t-1}$	0.9352	0.0583	0.0000***
TOTA	-5.5875	8.3126	0.5079
$TOTA_{t-1}$	22.6738	8.9787	0.0186**
FDI	5.8570	3.8732	0.1435
FDI_{t-1}	-7.9203	4.9582	0.1233
FDI_{t-2}	12.8735	3.9158	0.0031***
GEX	3.3171	2.7872	0.2457
GEX_{t-1}	-0.8938	2.9725	0.7663
GEX_{t-2}	-5.6087	2.7613	0.0534*
INF	0.2145	0.4449	0.6341
AG	-0.2999	2.1099	0.8882
AG_{t-1}	0.1540	2.6901	0.9548
AG_{t-2}	3.0499	1.8597	0.1141
Constant	-134.2850	166.9828	0.4292
R^2		0.9977	
Adjusted R^2		0.9965	
F-statistic		814.35 ($p < 0.001$)	
Observations		38 (1985–2022)	

Dependent variable: GDPPC. ARDL(1, 1, 2, 2, 0, 2) selected by AIC. *** p<0.01, ** p<0.05, * p<0.10.

The bounds test confirms a long-run relationship for the growth model (F = 6.876, above the 1% upper bound). However, the error correction term in Table 4.14 is small and insignificant (-0.0648, $p = 0.2772$), and none of the long-run coefficients is significant, so adjustment towards the long-run equilibrium is slow and imprecisely estimated. These results show that taxation has a delayed positive effect on economic growth. However, the poverty model provides no evidence that higher GDP per capita reduced poverty headcount, so there is no evidence that this growth effect translated into lower poverty during the study period, although this conclusion is limited by the interpolated poverty data.

Table 4.14: Long-Run Effects of Taxation on GDP per Capita

Variable	Coefficient	Std. Error	t-statistic	p-value
TOTA	263.4963	215.3642	1.223	0.2330
FDI	166.7095	174.4902	0.955	0.3489
GEX	-49.1241	41.7712	-1.176	0.2511
INF	3.3074	7.2838	0.454	0.6539
AG	44.7853	69.3225	0.646	0.5244
Constant	-2070.8819	4201.5684	-0.493	0.6266
ECT_{t-1} (speed of adjustment)	-0.0648	0.0583	-1.112	0.2772

Dependent variable: GDPPC. ARDL(1, 1, 2, 2, 0, 2); observations: 38 (1985–2022). *** p<0.01, ** p<0.05, * p<0.10.

4.6 Diagnostic Tests

Table 4.15 presents the diagnostic tests for each of the five estimated models. The Breusch–Godfrey LM test shows no evidence of serial correlation in any model, as all p-values exceed 0.05 (the lowest is 0.1436, for the poverty model). The Breusch–Pagan test shows no evidence of heteroskedasticity at the 5% level in any model, although heteroskedasticity is marginally significant at the 10% level in the income inequality model ($p = 0.0502$) and the growth model ($p = 0.0902$). The Jarque–Bera test indicates that the residuals of the income inequality, healthcare, and growth models are normally distributed (the lowest p-value is 0.1928), but normality is rejected for the education and poverty models ($p < 0.001$). The non-normal residuals in these two models are likely caused by the interpolated values, which create abrupt changes in slope at the survey years. With 38 observations, non-normality can affect the accuracy of p-values, so the results of these two models should be interpreted with additional caution. Otherwise, the diagnostic results support the reliability of the standard errors and hypothesis tests reported above.

Table 4.15: Diagnostic Tests for the Estimated ARDL Models

Diagnostic Test	**GINI**	**SSER**	**IMU**	**POVH**	**GDPPC**
Breusch–Godfrey Serial Correlation LM Test					
LM Statistic	0.0740	0.2263	0.0115	2.1394	0.6620
p-value	0.7856	0.6343	0.9145	0.1436	0.4159
Breusch–Pagan Heteroskedasticity Test					
BP Statistic	16.9078	9.0930	19.0752	11.7783	20.2048
p-value	0.0502*	0.4287	0.1208	0.3002	0.0902*
Jarque–Bera Normality Test					
JB Statistic	3.2920	20.3265	2.2762	29.9870	0.3155
p-value	0.1928	0.0000***	0.3204	0.0000***	0.8540
CUSUM within 5% bounds	No	Yes	No	Yes	Yes

Figure 4.5 presents the cumulative sum (CUSUM) stability tests, with the 5% significance bounds shown as dashed lines. The CUSUM plots for the education, poverty, and growth models remain within the bounds throughout the sample period, indicating that their coefficients are stable over time. However, the CUSUM plots for the income inequality model and the healthcare model cross the lower bound, from 2009 and from 2005 respectively, indicating

parameter instability. For the healthcare model, the timing coincides with the nationwide roll-out of the National Health Insurance Scheme, which changed how health services in Ghana are financed. For the income inequality model, it suggests that the relationship between inequality and its determinants changed during the later part of the sample. The results of these two models should therefore be interpreted with caution, and future research could explicitly allow for structural breaks.

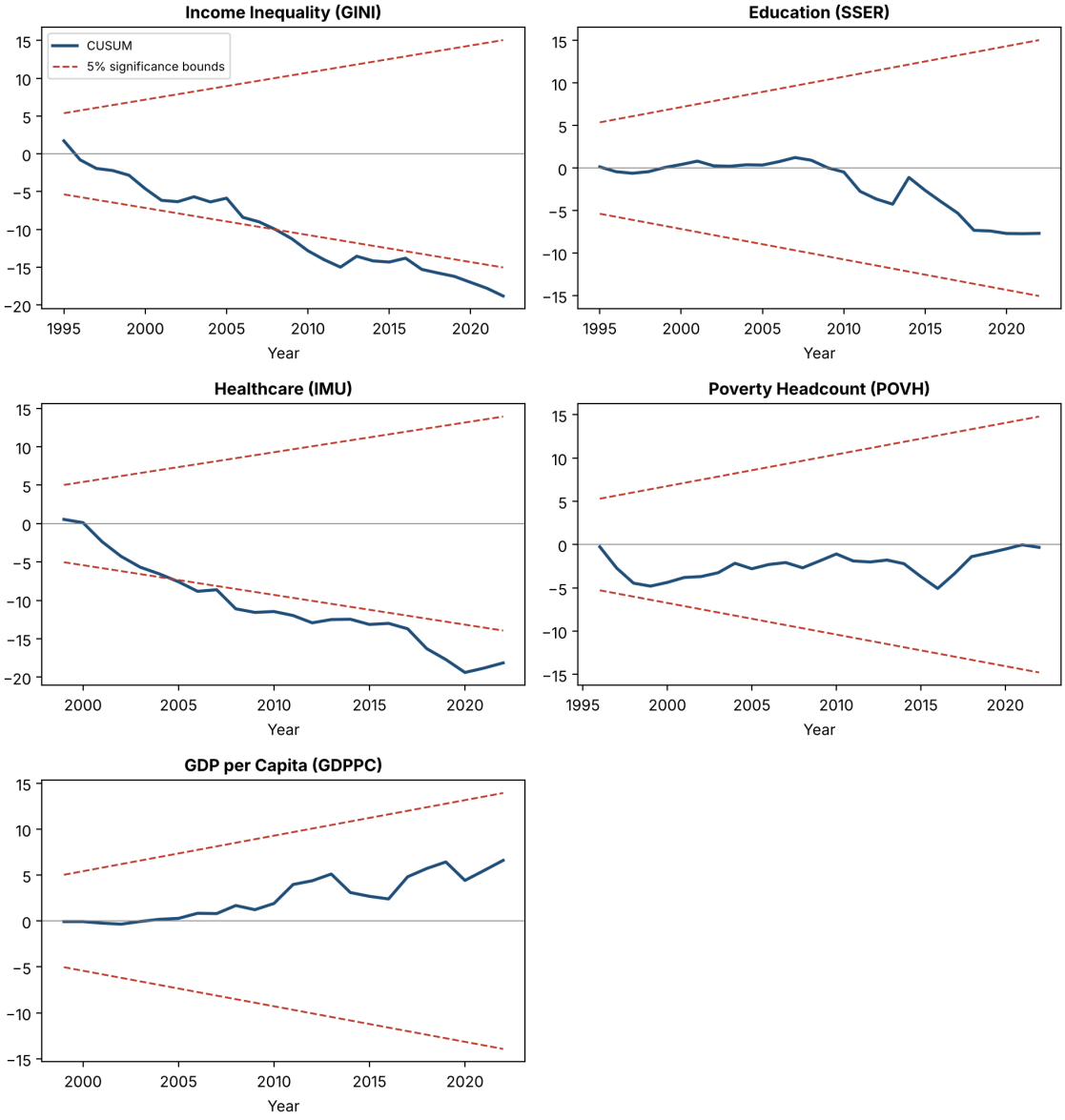


Figure 4.5: CUSUM Stability Tests for the Estimated ARDL Models (dashed lines: 5% significance bounds)

Chapter 5 Discussions

5.1 Introduction

This chapter presents a comprehensive discussion of the empirical findings reported in Chapter 4, interpreting them within the context of the study's objectives, theoretical frameworks, and existing empirical literature. The discussion is organized according to the four specific research objectives: (1) the effect of taxes on income inequality, (2) the effect of taxes on access to education, (3) the effect of taxes on access to healthcare, and (4) the effect of taxes on poverty reduction in Ghana. Each section synthesizes the quantitative results with theoretical expectations and prior studies, drawing implications for policy and future research.

5.2 Taxation and Income Inequality in Ghana

The ARDL estimates revealed that neither direct taxes nor indirect taxes exert statistically significant effects on income inequality in Ghana, whether in the short run, with lags, or in the long run. The short-run coefficients of direct taxes (0.0554, $p = 0.1460$) and indirect taxes (0.0486, $p = 0.3041$) were insignificant, as were their lags, despite strong persistence in the Gini coefficient itself (first lag = 1.6616, second lag = -0.7007, both $p < 0.001$). The bounds test found no long-run relationship ($F = 2.032$), and the long-run coefficients of direct taxes ($p = 0.1398$) and indirect taxes ($p = 0.7502$) were also insignificant, confirming the absence of systematic distributional effects from taxation.

The finding that Ghana's tax system does not significantly influence income inequality contradicts the **Ability-to-Pay Theory** (Jhingan, 2014), which posits that progressive taxation should redistribute income from higher to lower earners. Under this theoretical framework, direct taxes—particularly personal income tax with progressive rates (0% to 35% in Ghana)—should theoretically reduce inequality by extracting proportionally more from high-income households. Similarly, the **Benefit Theory** (Ayodeji & Others, 2021) would predict that tax-financed public services should indirectly reduce inequality by providing equalizing access to education, healthcare, and infrastructure. Several theoretical explanations may account for this unexpected finding. First, the **Theory of Realization and Exigent Implementation** (A. Adegbite, 2021) emphasizes that taxation must yield tangible benefits to taxpayers. In Ghana's context, the weak linkage between tax payments and visible public service improvements may undermine the redistributive potential of the tax system. Second, the substantial informal sector—estimated to constitute a significant portion of Ghana's economy—operates largely outside the formal tax net (Aringo, 2023; Anim et al., 2020). This structural characteristic means that many high-income earners in the informal sector may escape direct taxation entirely, limiting the progressivity of the tax system. The findings align with Obaretin et al. (2017), who used a 34-year time series dataset to examine taxation's influence on the Gini coefficient in Nigeria and found that direct, indirect, and total taxes did not significantly affect income inequality. Their conclusion that the Nigerian tax system may not adequately function as a redistributive tool resonates with the Ghanaian context documented in this study. Similarly, Omesi and Appah (2021) found that while personal and company income taxes exhibited significant negative relationships with inequality in Nigeria, petroleum profit tax showed positive associations, and value-added tax demonstrated insignificant effects—suggesting that the redistributive capacity of tax systems varies substantially across tax types and contexts. However, the results diverge from Xue (2018), who reported strong positive relationships between personal income tax and petroleum profit tax revenues and improved income redistribution in Nigeria. This divergence may reflect differences in tax structure, administrative capacity, or the composition of tax revenues between Ghana and Nigeria. The contrasting findings underscore the context-dependent nature of tax-redistribution linkages emphasized in the literature (Piketty & Saez, 2007; Palić et al., 2019).

The first research objective sought "to assess the effect of taxes on income inequality in Ghana."

The evidence indicates that Ghana's tax system, in its current configuration and administrative context, does not significantly influence income distribution. This finding has profound implications: taxation in Ghana appears to function primarily as a revenue-generating mechanism rather than a redistributive instrument. The progressive rate structure enshrined in the Income Tax Act, 2015 (Act 896) may be undermined by limited coverage, evasion, exemptions, and the predominance of regressive indirect taxes such as VAT (15%) in the revenue mix (Ghana Revenue Authority, 2023). Achieving redistributive objectives may require not only progressive rate structures but also expanded formal sector coverage, improved compliance, and strategic allocation of revenues to pro-poor expenditures.

5.3 Taxation and Access to Education in Ghana

The education model revealed a delayed pattern. Contemporaneously, direct taxes showed a positive and marginally significant relationship with secondary school enrollment (coefficient = 2.4631, p = 0.0790), and the first lag was negative and marginally significant (-3.2677, p = 0.0645). The second lag of direct taxes was positive and highly significant (4.6006, p = 0.0015): a one percentage point increase in direct taxes as a share of GDP is associated with a 4.60 percentage point increase in enrollment two years later. The bounds test was inconclusive (F = 4.289) and the error correction term was close to zero, so the effect of taxation on education operates through delayed short-run adjustments rather than a long-run equilibrium.

The delayed positive effect of direct taxes on secondary school enrollment strongly supports **Human Capital Theory** (Zimmer, 2022; Jiang & Shi, 2023), which emphasizes that investments in knowledge and skills—financed through public revenues—enhance individual productivity and societal prosperity. The two-year lag before positive effects materialize is theoretically consistent with fiscal implementation processes: tax revenues must be collected, appropriated through budgetary processes, allocated to the education sector, and translated into tangible improvements in school infrastructure, teacher recruitment, and educational materials before enrollment effects can be observed. The negative first lag, which is significant only at the 10% level, may point to a short-term trade-off consistent with the **Theory of Infrastructure-led Development** (Buthelezi, 2021). In the immediate term, increased tax collections may reduce

household disposable income, potentially creating liquidity constraints that delay educational enrollment decisions—particularly among low-income households where education costs, even in nominally "free" systems, include uniforms, supplies, and opportunity costs of forgone child labor. This short-term private cost must be incurred before the long-term public benefits of tax-financed educational investments materialize. The strong persistence in secondary school enrollment (first lag coefficient = 0.6332, $p < 0.001$) reflects the cumulative nature of educational progression, consistent with the **Endogenous Growth Model** predictions that human capital accumulation exhibits strong path dependence (Rothomi & Rafid, 2023). Once students enroll, they naturally continue through sequential grade levels, and educational infrastructure investments exhibit considerable durability. These findings align closely with A. Olarinde and Arike (2022), who investigated taxation and education financing in Nigeria using cointegration analysis and Vector Error Correction Models, finding a positive and statistically significant relationship between taxation and educational financing. Their conclusion that increased tax revenue allows governments to improve education services, expand access, and enhance quality is echoed in the delayed positive effect documented in this Ghanaian study. The results also support Oluyebo (2022), who analyzed petroleum profit tax and education tax in Nigeria, finding positive correlations with tertiary school enrollment. However, Oluyebo's observation that education tax revenue displayed a negative relationship with tertiary enrollment—potentially reflecting inefficiencies in collection or allocation—parallels the negative first lag documented here, suggesting that initial adjustment costs or implementation challenges may temporarily offset the positive effects of educational taxation. The findings extend Ayangbah and Sun (2017) recommendation that government grants should be increased to cover rising tertiary education costs by providing empirical evidence that tax-financed educational investments do, over time, translate into improved enrollment outcomes. The two-year adjustment pattern identified in this study offers a more precise temporal framework for policy expectations than previously available in the Ghanaian context.

The second research objective sought "to determine the effect of taxes on access to education in Ghana." The evidence demonstrates that direct taxes positively influence secondary school enrollment, but with a delayed response pattern, with a significant positive effect emerging approximately two years after taxes are collected. This finding validates the fiscal foundations of education policy while highlighting the importance of patience in evaluating tax-financed

educational interventions. Policymakers should not expect immediate enrollment responses to tax increases; rather, sustained revenue mobilization over multiple fiscal cycles is required to generate measurable improvements in educational access. The strong persistence in enrollment further implies that policy interventions, once effective, generate durable improvements through cumulative progression effects.

5.4 Taxation and Access to Healthcare in Ghana

The healthcare model, which uses immunization coverage as a proxy for access to healthcare, was the only welfare model with clear evidence of a long-run relationship with taxation. The bounds F-statistic (7.013) exceeded the upper bound at the 1% level, and the error correction term (-0.3754, $p < 0.001$) indicated that about 37.5% of any deviation from equilibrium is corrected each year. In the long run, indirect taxes significantly increased immunization coverage (5.5583, $p = 0.0337$), while direct taxes, non-tax revenue, and inflation were insignificant. In the short run, indirect taxes had a significant positive effect after two years (2.9003, $p = 0.0465$). However, the CUSUM test indicated parameter instability from 2005, which coincides with the nationwide roll-out of the NHIS, so these findings should be interpreted with appropriate caution.

The strong positive long-run relationship between taxation and healthcare access provides compelling support for the **Theory of Realization and Exigent Implementation** (A. Adegbite, 2021) in the health sector. This theory emphasizes that tax revenues should directly improve access, quality, and equity of medical services—particularly for marginalized populations. Ghana's National Health Insurance Scheme (NHIS), implemented in 2003 and financed through the National Health Insurance Levy (2.5% VAT) and Social Security contributions, exemplifies this theoretical mechanism: hypothecated taxes directly fund healthcare access, creating a visible linkage between tax payments and service benefits (Agyepong et al., 2016). The findings also align with the **Human Capital Theory** (Kucharska, 2022), which conceptualizes health as an integral component of human capital. Tax-financed health investments enhance population health status, which in turn increases labor productivity, earning capacity, and economic output—creating virtuous cycles of human development. The size of the long-run coefficient,

about 5.6 percentage points of immunization coverage for each percentage point of GDP raised through indirect taxes, suggests that the effect of tax revenue on healthcare access is economically meaningful, underscoring the importance of sustained health sector financing. These results corroborate A. Adegbite (2021), who examined taxation and health services in Nigeria using cointegration and VECM approaches, finding that taxation significantly influences healthcare provision in the long run. Adegbite's conclusion that tax revenues enable health infrastructure development, medical personnel training, and service expansion is strongly supported by the Ghanaian evidence. The findings also align with Adisa-Atta (2017), who studied public attitudes toward higher taxation for healthcare in Ghana, finding that only 35% of respondents supported higher taxes despite widespread healthcare access challenges. The positive long-run relationship documented in this study suggests that even with limited initial public support, sustained tax-financed health investments ultimately expand access—potentially building public confidence over time as the NHIS demonstrates tangible benefits. However, the results diverge from Doherty (2014), who found that increases in tax revenue do not automatically translate into higher government health expenditure in South Africa due to broader fiscal and economic policies. The stronger linkage observed in Ghana may reflect the hypothecated nature of the National Health Insurance Levy, which creates a dedicated revenue stream for health sector financing—a policy design feature that strengthens the tax-healthcare linkage compared to general revenue financing.

The third research objective sought "to determine the effect of taxes on access to healthcare in Ghana." The evidence indicates that taxation, specifically indirect taxation, exerts a positive and significant long-run effect on access to healthcare, as measured by immunization coverage. This finding supports continued reliance on tax financing, particularly the earmarked National Health Insurance Levy, for universal health coverage. The parameter instability detected around 2005 suggests that future research should explicitly model structural breaks associated with NHIS implementation and other health-sector reforms.

5.5 Taxation and Poverty Reduction in Ghana

The poverty headcount model revealed that direct taxes have no significant effect on poverty (-0.5049, $p = 0.1516$). Government expenditure had a negative and significant effect on poverty headcount (-0.1172, $p = 0.0295$), although the effect is modest in size. Poverty showed very strong persistence (first lag $= 0.9937$, $p < 0.001$). These results must be interpreted with caution: poverty headcount is observed in only seven GLSS survey years, the interpolated series behaves as an I(2) variable, so the bounds test is not valid, and the error correction term was close to zero (-0.0063, $p = 0.8003$), so no reliable long-run relationship could be established. The significant contemporaneous poverty-reducing effect of government expenditure strongly supports the **Social Welfare and Taxation for Poverty Reduction** framework (Maniquet & Neumann, 2021), which emphasizes that tax-financed social protection programs directly alleviate deprivation. Ghana's Livelihood Empowerment Against Poverty (LEAP) program, Ghana School Feeding Program (GSFP), and NHIS exemplify this mechanism—tax revenues finance cash transfers, nutritional support, and health insurance that immediately improve household welfare (Kipo-Sunyehzi, 2021; Tette & Enos, 2020).

The strong persistence in poverty (first lag coefficient $= 0.9937$) partly reflects the smoothness of the interpolated poverty series, but it is also consistent with the structural nature of deprivation described by poverty trap theories (Ravallion, 2016). Households experiencing poverty face multiple intersecting constraints—limited assets, poor health, low education, social exclusion—that create self-reinforcing cycles requiring sustained intervention to escape. The modest size of the expenditure effect suggests that public spending, which includes many items that do not target the poor, reduces poverty only to a limited extent. The positive coefficient of GDP per capita, which is contrary to the growth–poverty literature, likely reflects the interpolated poverty data, which are held constant after the last survey in 2016 while GDP per capita continued to grow. Reverse causality may also play a role: governments may raise spending in response to rising poverty, which weakens the observed negative relationship between expenditure and poverty. This interpretation aligns with the **Theory of Realization and Exigent Implementation**—governments deploy fiscal resources precisely when welfare outcomes deteriorate, generating observational patterns that complicate causal inference. These findings

align closely with Adu et al. (2023), who examined tax systems, poverty, and inequality in Ghana using NARDL approaches, finding that tax systems exert asymmetric effects with indirect taxes exerting more significant impacts than direct taxes. Their conclusion that indirect taxes disproportionately affect poverty resonates with the insignificant direct tax coefficients documented in this study, suggesting that Ghana's poverty dynamics may be more sensitive to consumption taxes than income taxes. The results also support Lofgren and Cicowiez (2022), who studied taxation, infrastructure investment, growth, and poverty reduction in Zimbabwe, finding complex intertemporal relationships between fiscal policy and poverty outcomes. Their emphasis on distinguishing between tax incidence, revenue utilization, and poverty dynamics is strongly validated by the Ghanaian evidence.

The findings extend Leventi et al. (2017), who studied tax-benefit policies in seven European countries, finding that child benefits, housing support, and social assistance reduce poverty—but effectiveness varies by country context. The Ghanaian evidence confirms that context matters profoundly: in developing economy settings with large informal sectors and limited administrative capacity, the poverty-reducing effects of direct taxation may be substantially attenuated compared to advanced economy contexts. The supplementary growth model showed that total taxes have a positive and significant effect on GDP per capita after one year (22.6738, $p = 0.0186$). However, the poverty model provides no evidence that higher GDP per capita reduced poverty headcount, so there is no evidence that this growth effect translated into lower poverty during the study period.

The fourth research objective sought "to assess the effect of taxes on poverty reduction in Ghana." The evidence reveals that direct taxes do not significantly influence poverty headcount in either the short run or the long run, while government expenditure is associated with lower poverty, although these results are indicative only because poverty data are available for only seven survey years. This finding has profound policy implications: tax policy alone does not reduce poverty; rather, it is the utilization of tax revenues through well-targeted government spending that generates poverty reduction. The distinction between revenue mobilization and expenditure effectiveness is critical—increasing tax collections without ensuring pro-poor allocation and efficient service delivery will not automatically reduce poverty. The modest size of the expenditure effect underscores the importance of distinguishing between expenditure levels

and expenditure effectiveness, highlighting the need for improved public financial management and poverty targeting in fiscal policy design.

5.6 Synthesis and Integration

Synthesizing findings across the four welfare dimensions reveals important patterns. First, **temporal heterogeneity** is pronounced: education effects materialize with a two-year lag, healthcare effects emerge in the long run, inequality effects are persistently absent, and poverty responds to government spending rather than to taxes directly. This temporal variation underscores the importance of dynamic specifications in fiscal policy evaluation. Second, **dimensional specificity** is evident: taxation demonstrates clear benefits for human capital development (education and healthcare) but ambiguous effects on distributional outcomes (inequality and poverty). This pattern suggests that Ghana's tax system may be more effective at financing service provision than at directly altering income distributions—a finding consistent with the revenue-focused orientation of tax administration. Third, the **primacy of government expenditure** emerges consistently: expenditure effects dominate tax effects across multiple specifications, particularly for poverty reduction. This finding validates the fiscal transmission mechanism—taxes enable spending, but spending produces welfare outcomes. Policy designs that strengthen the tax-expenditure linkage, such as hypothecated revenues for social programs, may enhance welfare effectiveness.

The findings contribute to theoretical development in several ways. The **Ability-to-Pay Theory** receives qualified support: progressive tax structures alone do not guarantee redistributive outcomes absent effective administration and broad coverage. The **Human Capital Theory** is strongly validated: tax-financed educational and health investments generate measurable improvements in human development, with a delayed effect on education and a long-run effect on healthcare. The **Theory of Realization and Exigent Implementation** is supported by the poverty-expenditure linkage: governments must not only collect revenues but deploy them effectively to generate welfare improvements. The **Social Welfare and Taxation for Poverty Reduction** framework is validated by the significant contemporaneous expenditure effects on poverty.

The mixed integration order of the variables (I(0) and I(1) once taxes are measured as a share of GDP, apart from the interpolated poverty headcount) confirmed the appropriateness of ARDL bounds testing for all models except the poverty model. All models passed the tests for serial correlation and heteroskedasticity (at the 5% level), but the residuals of the education and poverty models, which use interpolated data, were not normally distributed. The CUSUM tests confirmed parameter stability for the education, poverty, and growth models, but indicated instability in the income inequality and healthcare models, whose results should therefore be interpreted with caution. Despite these caveats, the diagnostic results support the reliability of the findings for policy inference.

Chapter 6 Conclusions and Recommendations

6.1 Introduction

This chapter presents the conclusions and recommendations derived from the empirical investigation into the impact of taxation on welfare in Ghana. The study was guided by four specific objectives: (1) to assess the effect of taxes on income inequality, (2) to determine the effect of taxes on access to education, (3) to determine the effect of taxes on access to healthcare, and (4) to assess the effect of taxes on poverty reduction. The chapter begins with a brief summary of the study and methodology, followed by conclusions organized according to each research objective. Policy recommendations are then proposed, and the chapter concludes with suggestions for future research.

6.2 Conclusions

This study examined the impact of taxation on welfare outcomes in Ghana over the period 1983 to 2022. The research adopted a positivist philosophy and employed a quantitative research design using time series data. Data were sourced from the Ghana Statistical Service StatsBank, the World Bank World Development Indicators database, and the UNU-WIDER Government Revenue Dataset, with all tax variables measured as a percentage of GDP. The study utilized the Autoregressive Distributed Lag (ARDL) bounds testing approach to cointegration, which

was appropriate given the mixed integration order of the variables (I(0) and I(1)) and the relatively small sample size. Four econometric models were specified to examine the relationships between tax variables (direct taxes, indirect taxes, and total tax revenue) and welfare indicators (income inequality measured by the Gini coefficient, access to education measured by secondary school enrollment, access to healthcare measured by immunization coverage, and poverty reduction measured by poverty headcount). Diagnostic tests, including tests for serial correlation, heteroskedasticity, normality, and parameter stability (CUSUM), were conducted to validate the robustness of the findings.

The study concludes that Ghana's tax system does not significantly influence income inequality. Neither direct taxes nor indirect taxes demonstrated statistically significant effects on the Gini coefficient in either the short run or with lagged adjustments, and the bounds test found no long-run relationship between taxation and inequality. This finding indicates that taxation in Ghana functions primarily as a revenue-generating mechanism rather than an instrument for income redistribution. The progressive rate structure enshrined in the Income Tax Act, 2015 (Act 896) appears to be undermined by limited coverage of the formal sector, significant tax evasion, the predominance of regressive indirect taxes such as VAT (15%) in the revenue mix, and the large informal sector that operates largely outside the tax net. Consequently, the redistributive potential of Ghana's tax system remains unrealized, and income inequality persists independently of fiscal interventions.

The study concludes that direct taxes positively influence access to education in Ghana, as measured by secondary school enrollment, but with a distinct delayed response pattern. Contemporaneously, direct taxes showed no significant effect on enrollment. However, direct taxes had a significant positive effect two years after collection: a one percentage point increase in direct taxes as a share of GDP was associated with a 4.60 percentage point rise in enrollment. This pattern reflects the fiscal implementation process: tax revenues must be collected, appropriated through budgetary processes, allocated to the education sector, and translated into tangible improvements in school infrastructure, teacher recruitment, and educational materials before enrollment effects materialize. The strong persistence in secondary school enrollment confirms the cumulative nature of educational progression. Although no stable long-run relationship was found, the delayed positive effect demonstrates that tax-financed educational investments

expand access to education over time, validating the fiscal foundations of Ghana's educational policies including the Free Compulsory Universal Basic Education (FCUBE) program and the Free Senior High School policy.

The study concludes that indirect taxation exerts a positive and significant long-run effect on access to healthcare in Ghana. The healthcare model, which measures access by immunization coverage, was the only welfare model with a clear long-run relationship with taxation: a one percentage point increase in indirect taxes as a share of GDP raised immunization coverage by about 5.6 percentage points in the long run, and about 37.5% of any deviation from equilibrium was corrected each year. This finding validates the welfare state hypothesis and supports the continued reliance on tax financing for universal health coverage initiatives. The National Health Insurance Scheme (NHIS), financed through the National Health Insurance Levy (a component of VAT) and Social Security contributions, exemplifies the mechanism through which tax revenues directly fund healthcare access, creating a visible linkage between tax payments and service benefits. The hypothecated nature of health financing in Ghana appears to strengthen the tax-healthcare linkage compared to general revenue financing, providing a model for other social sectors. While the CUSUM test indicated parameter instability in the healthcare model from 2005, when the NHIS was rolled out nationwide, the positive long-run effect of indirect taxes is theoretically consistent with Ghana's health financing arrangements.

The study concludes that direct taxes do not significantly influence poverty reduction in Ghana, while government expenditure is associated with significant reductions in poverty. The poverty headcount model revealed that government expenditure significantly reduced poverty (coefficient = -0.1172, p = 0.0295), while direct taxes were insignificant. This finding has profound implications: tax policy alone does not reduce poverty; rather, it is the utilization of tax revenues through well-targeted government spending that generates poverty reduction. Ghana's social protection programs—including the Livelihood Empowerment Against Poverty (LEAP) program, the Ghana School Feeding Program (GSFP), and the National Health Insurance Scheme (NHIS)—exemplify the mechanisms through which tax-financed expenditures directly improve household welfare. These poverty results should, however, be interpreted with caution, because poverty headcount is observed in only seven survey years and the interpolated series behaves as an I(2) variable, so no reliable long-run relationship could be estimated.

6.3 Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed:

6.3.1 Recommendations for Tax Policy and Income Inequality

1. **Broaden the Tax Base to Include the Informal Sector:** The Ghana Revenue Authority should intensify efforts to integrate the substantial informal sector into the formal tax net. This can be achieved through simplified presumptive tax regimes, digital registration systems, and taxpayer education programs that emphasize the link between tax payments and public services. Broadening the base will enhance the progressivity and redistributive potential of the tax system.

2. **Strengthen Progressive Elements of the Tax System:** Policymakers should review the rate structure of direct taxes to ensure genuine progressivity, including regular adjustments of tax brackets for inflation to prevent fiscal drag. Simultaneously, consideration should be given to reducing reliance on regressive indirect taxes such as VAT, or introducing compensatory mechanisms such as cash transfers to offset the burden on low-income households.

3. **Enhance Tax Compliance and Enforcement:** The Ghana Revenue Authority should invest in modern tax administration technologies, including data analytics and third-party information matching, to reduce evasion and improve compliance. Strengthened enforcement will ensure that those with greater ability to pay actually contribute their fair share, enhancing the redistributive capacity of the tax system.

6.3.2 Recommendations for Tax Policy and Education

4. **Maintain and Strengthen Education-Specific Tax Financing:** Given the delayed positive effect of direct taxation on educational access, policymakers should maintain dedicated education financing mechanisms, including the District Assembly Common Fund

and education-specific levies. The positive two-year lag effect suggests that sustained, predictable education financing is essential for enrollment improvements.

5. **Stabilize Education Funding to Smooth Adjustment Costs:** The negative (though statistically insignificant) first-year coefficient on enrollment following tax increases suggests that households face short-term liquidity constraints. Policymakers should consider compensatory mechanisms during tax transitions, such as expanded textbook subsidies, school feeding programs, or transportation support, to mitigate immediate enrollment disruptions.

6. **Link Tax Revenue to Visible Educational Improvements:** To build public support for taxation and strengthen the tax-education linkage, the Ministry of Education should clearly communicate how tax revenues translate into educational improvements—new classrooms, trained teachers, learning materials—reinforcing the Benefit Theory and enhancing tax morale.

6.3.3 Recommendations for Tax Policy and Healthcare

7. **Protect and Expand Hypothecated Health Financing:** The positive long-run effect of indirect taxes on healthcare access documented in this study supports the continued use of hypothecated taxes for health financing, such as the National Health Insurance Levy. Policymakers should protect these dedicated revenue streams from diversion and consider expanding them to achieve universal health coverage.

8. **Strengthen the NHIS to Enhance Access:** Additional tax revenues allocated to the National Health Insurance Scheme should be used to expand coverage to excluded populations, reduce waiting times, improve service quality, and ensure that the scheme's benefits are tangible and visible to contributors. Visible health improvements will reinforce the tax-health linkage and build public confidence.

9. **Investigate Structural Change in Health Financing:** The parameter instability detected in the healthcare model around 2005, when the NHIS was rolled out nationwide, suggests that the relationship between taxation and health access may be more complex

than currently modeled. The Ministry of Health should commission further research to understand structural breaks, policy thresholds, and nonlinearities in the tax-health relationship.

6.3.4 Recommendations for Tax Policy and Poverty Reduction

10. **Strengthen the Tax-Expenditure-Poverty Linkage:** Since poverty reduction operates through government expenditure rather than taxation directly, policymakers should ensure that tax revenues are efficiently channeled into pro-poor programs. The LEAP program, GSFP, and NHIS should be adequately funded and continuously evaluated for targeting effectiveness.

11. **Improve Public Expenditure Efficiency and Targeting:** The modest size of the expenditure effect on poverty underscores the importance of distinguishing between expenditure levels and expenditure effectiveness. The Ministry of Finance should strengthen public financial management systems, enhance monitoring and evaluation, and ensure that poverty-targeted programs reach intended beneficiaries with minimal leakage.

12. **Integrate Tax and Expenditure Policy Design:** Tax and expenditure policies should be designed jointly to maximize welfare impacts. For example, expanding VAT while simultaneously increasing LEAP transfers can offset regressive effects while maintaining revenue adequacy. Integrated fiscal incidence analysis should inform policy decisions.

13. **Protect Social Spending During Fiscal Consolidation:** The significant contemporaneous poverty-reducing effects of government expenditure imply that social spending should be protected during periods of fiscal adjustment. Austerity measures that reduce social program funding risk reversing poverty gains and undermining long-term human capital development.

6.3.5 General Recommendations

14. **Strengthen Tax Administration Capacity:** Across all objectives, the effectiveness of tax policy depends on administrative capacity. Continued investment in Ghana Revenue Authority modernization, staff training, and digital infrastructure is essential for realizing the welfare potential of taxation.

15. **Enhance Data Collection and Availability:** Future policy analysis and research require improved data on tax incidence, program coverage, and welfare outcomes. The Ghana Statistical Service should prioritize the collection and dissemination of high-quality, timely data to support evidence-based policymaking.

16. **Promote Policy Coherence Across Sectors:** Welfare outcomes are shaped by interactions among tax policy, expenditure policy, and broader economic conditions. Sectoral ministries should coordinate policy design to ensure that tax, education, health, and social protection policies work synergistically rather than at cross-purposes.

6.4 Contributions to Knowledge

This study makes several contributions to the literature on taxation and welfare in developing economies:

1. **Context-Specific Empirical Evidence:** The study provides robust, context-specific empirical evidence on tax-welfare linkages in Ghana, addressing a significant research gap identified in the literature review. Prior studies focused predominantly on Nigeria or cross-country comparisons, leaving Ghana under-researched.

2. **Temporal Specificity:** By employing ARDL modeling with lag structures, the study identifies precise temporal patterns in tax effects—the two-year lag for education, the long-run horizon for healthcare, and the immediate expenditure effects for poverty—providing novel insights for policy timing expectations.

3. **Multi-Dimensional Welfare Assessment:** The study examines four distinct welfare dimensions within a unified analytical framework, revealing heterogeneous effects across dimensions and highlighting the importance of dimension-specific policy design.

4. **Tax-Expenditure Distinction:** The findings underscore the critical distinction between tax mobilization and expenditure effectiveness, demonstrating that welfare outcomes depend on both revenue adequacy and spending efficiency—a distinction often blurred in policy discourse.

5. **Methodological Rigor:** The application of ARDL bounds testing with comprehensive diagnostic verification provides a methodological template for future fiscal policy research in developing economy contexts with mixed integration properties and limited sample sizes.

6.5 Limitations of the Study

The following limitations should be considered when interpreting the findings:

1. **Data Constraints:** The study relied on secondary data with potential limitations in accuracy, consistency, and frequency. Some welfare indicators, particularly healthcare access measures, may not fully capture the multidimensional nature of healthcare access.

2. **Aggregation of Tax Variables:** Tax variables were analyzed at aggregate levels (direct, indirect, total). Disaggregated analysis of specific tax types (personal income tax, corporate tax, VAT, excise duties) might reveal heterogeneous effects masked by aggregation.

3. **Measurement of Welfare Outcomes:** Welfare indicators such as secondary school enrollment capture access but not quality. Future research should examine tax effects on educational quality, learning outcomes, and health status rather than merely access.

4. **Causality Limitations:** While ARDL modeling with lags strengthens causal inference, the observational nature of the data limits definitive causal conclusions. Unobserved confounders may influence estimated relationships.

5. **Interpolated Data:** The Gini coefficient and poverty headcount are observed in only seven GLSS survey years, and secondary school enrollment is missing for eight years; these gaps were filled by linear interpolation. The interpolated series are smoother than the true series and contain less independent information than 40 annual observations would suggest, and the interpolated poverty headcount behaves as an I(2) series, so the poverty results are indicative only. Future research could use an annual poverty proxy, such as household final consumption expenditure per capita, which does not require interpolation.

6. **Parameter Stability and Sample Size:** The CUSUM tests indicated parameter instability in the income inequality and healthcare models, and the estimation sample of 38 annual observations limits the precision of the long-run estimates. Immunization coverage is also only one dimension of access to healthcare.

6.6 Suggestions for Future Research

Based on the findings and limitations of this study, the following directions for future research are proposed:

1. **Disaggregated Tax Analysis:** Future studies should disaggregate tax variables to examine the differential effects of personal income tax, corporate tax, VAT, excise duties, and trade taxes on welfare outcomes.

2. **Quality Dimensions:** Research should extend beyond access measures to examine tax effects on educational quality (learning outcomes, completion rates) and healthcare quality (health status, mortality rates, patient satisfaction).

3. **Nonlinear and Threshold Effects:** The NARDL approach should be employed to examine potential asymmetries in tax-welfare relationship to determine whether positive and negative tax shocks have symmetric effects.

4. **Micro-Level Analysis:** Household survey data should be utilized to examine tax incidence and welfare effects at the household level, capturing heterogeneity across income groups, regions, and demographic characteristics.

5. **Comparative Studies:** Cross-country comparative studies within West Africa would illuminate how institutional differences, tax structures, and development levels moderate tax-welfare relationships.

6. **Political Economy Analysis:** Qualitative research should explore the political economy factors shaping tax policy design, implementation, and welfare outcomes, including interest group dynamics, bureaucratic capacity, and citizen-state relations.

7. **Expenditure Composition Effects:** Future research should disaggregate government expenditure to examine which expenditure categories (education, health, social protection, infrastructure) most effectively translate tax revenues into welfare improvements.

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